

Chapter 2

Brownian Motion

Gaussian processes form a class of stochastic processes used in several branches in pure and applied mathematics. The most important Gauss process is the Brownian motion; we recall the definition of Brownian motion.

Definition 2.0.1 *A stochastic process $B = (B_t)_{t \geq 0}$ defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called a (standard) Brownian motion, if*

i) $B_0 = 0$ a.s.

ii) for all $n \in \mathbb{N}$ and $0 \leq t_0 < t_1 < \dots < t_n$, we have

$$(B_{t_1} - B_{t_0}, \dots, B_{t_n} - B_{t_{n-1}}) \sim \mathcal{N}(0, \Sigma_{t_1, \dots, t_n}), \quad \Sigma_{t_1, \dots, t_n} := \begin{pmatrix} t_1 & & & 0 \\ & t_2 - t_1 & & \\ & & \ddots & \\ 0 & & & t_n - t_{n-1} \end{pmatrix}$$

i.e. the increments $\Delta B_{t_k} := B_{t_k} - B_{t_{k-1}}, k = 1, \dots, n$ are independent and $\mathcal{N}(0, t_k - t_{k-1})$ -distributed;

iii) the sample paths $t \mapsto B_t(\omega)$ are continuous $\mathbb{P}$ -almost surely.

The second property says that Brownian motion has stationary, normally distributed, independent increments. In what follows we provide three different constructions of Brownian motion.

2.1 Brownian motion as a Gauss processes

Definition 2.1.1 *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random variable $f : \Omega \rightarrow \mathbb{R}$ is called Gaussian if*

$$\mathbb{P}(f \in B) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_B e^{-\frac{1}{2\sigma^2}(x-m)^2} dx, \quad B \in \mathcal{B}$$

for some $m \in \mathbb{R}$ and $\sigma > 0$. A random vector

$$f = (f_1, \dots, f_n) : \Omega \rightarrow \mathbb{R}^n$$

is called Gaussian if

$$\langle f, a \rangle := \sum_{i=1}^n a_i f_i$$

is Gaussian for all $a = (a_1, \dots, a_n) \in \mathbb{R}^n$. We call

$$m = (m_1, \dots, m_n), \quad m_i := \mathbb{E}f_i$$

the mean vector and

$$\sigma = (\sigma_{ij}), \quad \sigma_{ij} := \mathbb{E}[(f_i - m_i)(f_j - m_j)]$$

the covariance (matrix). If $n = 1$, then we put $m = \mathbb{E}f$ and $\sigma^2 = \mathbb{E}(f - m)$.

We recall that probability measures are uniquely determined by their characteristic functions and that the characteristic function of a d -dimensional Gaussian random variable X with mean vector m and covariance matrix C is given by

$$\phi_X(\theta) = \exp\left(i\langle \theta, m \rangle - \frac{1}{2}\theta^t C \theta\right), \quad \theta \in \mathbb{R}^d.$$

Exercise 16 Show that if f, g are Gaussian random variables with the same mean vector and covariance matrix, then they have the same distributions.

Definition 2.1.2 A stochastic process X is called Gaussian if its finite dimensional distributions are Gaussian. The mean process and the covariance function are given by, respectively,

$$m_t := \mathbb{E}X_t, \quad \Gamma(s, t) := \mathbb{E}[(X_t - m_t)(X_s - m_s)].$$

If B is Brownian motion, then $m_t = 0$ by definition and for $s \leq t$, by the independence of the increments,

$$\begin{aligned} \Gamma(s, t) &= \mathbb{E}[B_s B_t] \\ &= \mathbb{E}[B_s(B_t - B_s)] + \mathbb{E}B_s^2 \\ &= \mathbb{E}[B_s]\mathbb{E}[B_t - B_s] + s = s. \end{aligned}$$

Hence

$$\Gamma(s, t) = s \wedge t.$$

We know how to extend a consistent family of finite-dimensional distributions to a stochastic process from Stochastic II.

Theorem 2.1.3 (Daniel-Kolmogorov) Let S be a Polish space equipped with its Borel σ -field $\mathcal{S}$. Let

$$\{Q^J : J \subset I, |J| < \infty\}$$

be a consistent family of finite-dimensional distributions Q^J on $(S^J, \mathcal{S}^J)$. Then, there exists a unique probability measure $\mathbb{P}$ on $(S^I, \mathcal{S}^I)$ that has all the Q^J as its finite dimensional distributions.

Let $(S^I, \mathcal{S}^I, \mathbb{P})$ as in the above theorem and $\pi = (\pi_t)_{t \in I}$ be the canonical process defined by $\pi_t(\omega) = \omega_t$ for all $t \in I$ where $\omega = (\omega_t)_{t \in I}$. Then the finite dimensional distributions of π are given by the family of probability measures $\{Q^J : J \subset I, |J| < \infty\}$. With this, we are now ready to prove the existence of Gauss processes and to show that a Gauss process is uniquely (in distribution) determined by its covariance function.

Theorem 2.1.4 (*Existence of Gaussian processes*) Let

$$\Gamma : I \times I \rightarrow \mathbb{R}$$

be positive, semi-definite and symmetric, i.e.

$$\sum_{i,j=1}^n \Gamma(t_i, t_j) a_i a_j \geq 0, \quad \Gamma(s, t) = \Gamma(t, s)$$

for all $t, s, t_1, \dots, t_n \in I$, $a_1, \dots, a_n \in \mathbb{R}$ and $n \in \mathbb{N}$. Then, there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a Gaussian process X on it such that

$$\mathbb{E}X_t = 0, \quad \mathbb{E}X_s X_t = \Gamma(s, t)$$

PROOF: We construct a consistent family of probability measures with the desired properties. To this end, let γ_n be a standard Gauss measure on $\mathbb{R}^n$ and

$$C = (\Gamma(t_i, t_j))_{i,j=1}^n.$$

The matrix C is symmetric, positive semi-definite and by the Cholesky decomposition there exists a matrix A such that

$$C = AA^t.$$

Let $\mu_n := \gamma_n \circ A^{-1}$. Let e_i be the i th unit vector in $\mathbb{R}^n$. Since every linear combination of normal random variables is normal

$$\int \langle x, e_i \rangle \mu_n(dx) = \int \langle Ax, e_i \rangle \gamma_n(dx) = 0$$

and

$$\begin{aligned} \int \langle x, e_i \rangle \langle x, e_j \rangle \mu_n(dx) &= \int \langle Ax, e_i \rangle \langle Ax, e_j \rangle \gamma_n(dx) \\ &= \int \langle x, A^t e_i \rangle \langle x, A^t e_j \rangle \gamma_n(dx) \\ &= \langle A^t e_i, A^t e_j \rangle \\ &= \langle e_i, C e_j \rangle \end{aligned}$$

This defines a consistent family of Gaussian measures (exercise). Hence, there exists a unique probability measure ν on the measure space

$$(\Omega, \mathcal{F}) = (\mathbb{R}^I, \mathcal{B}^I)$$

such that the coordinate process $X_t(\omega) = \omega(t)$ has the desired properties. $\square$

Exercise 17 Prove that the family of measures defined in the previous proof is consistent.

Example 2.1.5 (*Brownian bridge*) Let $I = [0, 1]$. We want to construct a Gaussian process X that starts and ends in 0. One possibility is to choose a Brownian motion B and to define

$$X_t := B_t - tB_1, \quad t \in [0, 1];$$

we will see an alternative construction later. Then, for all $0 \leq t_1 \leq \dots \leq t_d < 1$,

$$(X_{t_1}, \dots, X_{t_d}) = A(B_{t_1}, \dots, B_{t_d}, B_1)^t, \quad A = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & -t_1 \\ 0 & 1 & 0 & \dots & 0 & -t_2 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & -t_d \end{pmatrix}.$$

Hence, X is a Gaussian process; its covariance function can easily be computed: for each $0 \leq s \leq t \leq 1$

$$\begin{aligned} \mathbb{E}[X_s X_t] &= \mathbb{E}[(B_s - sB_1)(B_t - tB_1)] \\ &= \mathbb{E}B_s B_t - t\mathbb{E}B_s B_1 - s\mathbb{E}B_t B_1 + st\mathbb{E}B_1^2 \\ &= s(1 - t). \end{aligned}$$

Example 2.1.6 (Fractional Brownian motion) There exists a unique Gauss process X with the following covariance function:

$$\Gamma(s, t) = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H})$$

where $H \in (0, 1)$ is the Hurst coefficient. The process is called fractional Brownian motion with Hurst coefficient H . The case $H = 1/2$ corresponds to Brownian motion. It is not entirely trivial to show that Γ satisfies the properties of a covariance function.

Up to now we have constructed stochastic processes with certain finite-dimensional distributions. In the case of Gaussian processes this can be done through the covariance structure. Unfortunately, we cannot guarantee that the probability measure on $(S^I, \mathcal{S}^I)$ constructed via Daniel-Kolmogorov is concentrated on the set of continuous functions. To the contrary, the set $C(I, \mathbb{R})$ of continuous real-valued functions on I is usually not an element of $\mathcal{S}^I$ as shown by the following lemma.

Lemma 2.1.7 • Let $(E, \mathcal{B})$ be a measurable space and I be a non-empty index set. For any set $B \in \mathcal{B}^I$ there exists a countable set $S \subset I$ such that for any pair $(x, y) \in E^I \times B$ the following implication holds:

$$x(t) = y(t) \text{ for all } t \in S \text{ implies } x \in B.$$

That is, any set of functions $B \in \mathcal{S}^I$ is determined by the values on a countable set $J \subset I$.

• Let E be a metric space that contains at least two elements and $\mathcal{B}$ be a σ -field on E . Then

$$C(\mathbb{R}_+, E) \notin \mathcal{B}^{\mathbb{R}_+}.$$

PROOF: To see the first assertion let us recall that the σ -field $\mathcal{B}^I$ is generated by the sets

$$\text{pr}_J^{-1}(B_0); \quad J \subset I \text{ finite}; \quad B_0 \in \mathcal{B}^J.$$

Let $\mathcal{B}^a$ be the class of all sets $B \subseteq E^I$ for which there exists a countable set $S \subseteq I$ such that, for any pair $x \in E^I$ and $y \in B$ the above implication holds. Then $\mathcal{B}^a$ is a σ -field. The properties of a σ -field are evident except, perhaps, stability w.r.t. complements, which can be seen as follows. Let $B \in \mathcal{B}^a$ and S be the corresponding countable index set. Let $x \in E^I$ and $y \in B^c$ with $x = y$ on S . Then $x \notin B$ as otherwise $y \in B$ (note that $y \in E^I$). Hence $x \in B^c$ and so $B^c \in \mathcal{B}^a$. Now, choosing $S = J$ we see that $\text{pr}_J^{-1}(B_0) \in \mathcal{B}^a$ for all $B_0 \in \mathcal{B}^J$. This shows that $\mathcal{B}^a = \mathcal{B}^I$.

As for the second assertion, let us assume to the contrary that $C(\mathbb{R}_+, E) \in \mathcal{B}^{\mathbb{R}^+}$ then there exists a countable set S such that any mapping x that coincides with $y \in C$ on S is continuous. Adding the set of rational numbers to S if necessary, we may assume that $S \subseteq \mathbb{R}_+$ dense and so y is uniquely determined by its values on S . Since E contains two points $e_1 \neq e_2$ we can now choose $t_0 \notin S$ such that

$$x(t_0) = e_1; \quad x(t) = e_2 \quad \text{for } t \neq t_0; \quad y \equiv e_2.$$

Then $x = y$ on S but $x \neq y$ on $\mathbb{R}_+$ a contradiction. $\square$

The following remark sheds additional light on the preceding result.

Remark 2.1.8 *Let E be a complete separable metric space equipped with the Borel- σ -field $\mathcal{E} = \mathcal{B}(E)$. If I is a countable set, then $\mathcal{E}^I = \mathcal{B}(E)^I = \mathcal{B}(E^I)$. The above lemma shows that this is not true for uncountable index sets. The intuitive reason is that σ -algebras are only stable w.r.t. countable but not uncountable operations.*

In view of the preceding lemma an additional argument is required when constructing continuous stochastic processes. The next theorem provides a sufficient condition that guarantees that a stochastic process has a continuous modification. In this case, we shall always work with this modification.

Theorem 2.1.9 (Kolmogorov-Centsov) *Let X be a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ such that*

$$\mathbb{E}[|X_t - X_s|^\alpha] \leq C|t - s|^{1+\beta}, \quad 0 \leq s \leq t \leq T$$

for some $\alpha, \beta, C > 0$. Then there exists a continuous modification $\tilde{X}$ that satisfies

$$\mathbb{P}\left(\sup_{0 < t-s < h(\omega)} \frac{|\tilde{X}_t(\omega) - \tilde{X}_s(\omega)|}{|t-s|^\gamma} \leq \delta\right) = 1$$

for some $\delta > 0$, where h is a positive random variable and $\gamma \in (0, \frac{\beta}{\alpha})$.

PROOF: Let us take $T = 1$. We establish “increasing degrees of continuity”.

- It follows from Tschebyscheff’s inequality and the moment condition on X that the mapping $t \rightarrow X_t(\omega)$ is continuous in probability:

$$\mathbb{P}(|X_t - X_s| \geq \varepsilon) \leq \frac{\mathbb{E}|X_t - X_s|^\alpha}{\varepsilon^\alpha} \leq C\varepsilon^{-\alpha}|t - s|^{1+\beta}.$$

- Next, we establish uniform continuity on a set of full measure. To this end, we apply the above inequality with $t := \frac{k}{2^n}$, $s := \frac{k-1}{2^n}$ and $\varepsilon = 2^{-\gamma n}$. Then

$$\mathbb{P}\left(|X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\right) \leq C2^{-n(1+\beta-\alpha\gamma)}.$$

Thus,

$$\mathbb{P}\left(\max_{1 \leq k \leq 2^n} |X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\right) = \mathbb{P}\left(\bigcup_{k=1}^{2^n} |X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\right) \leq C2^{-n(\beta-\alpha\gamma)}$$

By the Borel-Cantelli lemma (recalled below) this yields the existence of a set Ω^* of full measure such that for all $\omega \in \Omega^*$,

$$\max_{1 \leq k \leq 2^n} |X_{\frac{k}{2^n}}(\omega) - X_{\frac{k-1}{2^n}}(\omega)| \leq 2^{-\gamma n}, \quad n \geq n^*(\omega) \quad (2.1)$$

where n^* is some integer-valued random variable. In order to show uniform continuity on Ω^* , we set

$$D := \bigcup_{i=1}^{\infty} D_n; \quad D_n := \left\{ \frac{k}{2^n}, k = 0, \dots, 2^n \right\}$$

and fix $\omega \in \Omega^*$, $n \geq n^*(\omega)$, $m > n$ and $0 < t - s < 2^{-n}$. We show below that for all $t, s \in D_m$

$$|X_t(\omega) - X_s(\omega)| \leq 2 \sum_{j=n+1}^m 2^{-\gamma j}. \quad (2.2)$$

For all $s, t \in D$ with $0 < t - s < h(\omega) := 2^{-n^*(\omega)}$, we select $n > n^*(\omega)$ such that

$$2^{-(n+1)} \leq t - s < 2^{-n}.$$

For any such s, t we get from (2.2) that

$$|X_t(\omega) - X_s(\omega)| \leq 2 \sum_{j=n+1}^{\infty} 2^{-\gamma j} \leq 22^{-\gamma(n+1)} \sum_{j=0}^{\infty} 2^{-\gamma j} \leq \delta |t - s|^{\gamma}, \quad \delta := \frac{2}{1 - 2^{-\gamma}}. \quad (2.3)$$

- We now define the continuous modification. For $\omega \notin \Omega^*$ we set $\tilde{X}_t(\omega) \equiv 0$. For $\omega \in \Omega^*$ and $t \in D$ we set $\tilde{X}_t(\omega) = X_t(\omega)$. For $\omega \in \Omega^*$ and $t \in D^c$ we choose a sequence $\{s_n\} \subset D$ that converges to t . By the uniform continuity of $X(\omega)$ on D , the sequence $\{X_{s_n}(\omega)\}$ has a limit $\tilde{X}_t(\omega)$ that depends on t but not on the choice of the sequence $\{s_n\}$. The process $\tilde{X}$ satisfies almost surely the path-wise Hölder continuity (2.3). Moreover, since $\{X_{s_n}\}$ converges in probability to X_t and a.s. to $\tilde{X}_t$, we conclude that $X_t = \tilde{X}_t$ a.s.
- It remains to establish (2.2). We proceed by induction. For $m = n + 1$ the assertion is correct by the choice of Ω^* . Let us therefore assume it is also correct for $m = n + 1, \dots, M - 1$ and let $s < t, s, t \in D_M$. Let

$$t' := \max\{u \in D_{M-1} : u \leq t\}; \quad s' := \max\{u \in D_{M-1} : u \geq s\}.$$

Then $s \leq s' \leq t' \leq t$, and $s' - s < 2^{-M}$, $t - t' < 2^{-M}$. By (2.1)

$$|X_{s'}(\omega) - X_s(\omega)| \leq 2^{-\gamma M}, \quad |X_{t'}(\omega) - X_t(\omega)| \leq 2^{-\gamma M}$$

and by the induction hypothesis, since $t', s' \in D_{M-1}$,

$$|X_{t'}(\omega) - X_{s'}(\omega)| \leq 2 \sum_{j=n+1}^{M-1} 2^{-\gamma j}.$$

□

The proof of the preceding theorem used the Borell-Cantelli-Lemma.

Lemma 2.1.10 (Borell-Cantelli) *Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of events defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.*

- (i) *If $\sum_n \mathbb{P}[A_n] < \infty$, only finitely many A_n occur: $\mathbb{P}[\text{only finitely many } A_n \text{ occur}] = 1$.*

(ii) If $\sum_n \mathbb{P}[A_n] = \infty$ and the events $A_1, A_2, \dots$ are independent, infinitely many A_n occur:

$$\mathbb{P}[\text{infinitely many } A_n \text{ occur}] = 1.$$

Since the covariance function $\Gamma(s, t) = s \wedge t$ is symmetric and positively semi-definite, we know that there exists a centered Gaussian process with this covariance function. This process satisfies the following properties, all derived entirely(!) from the covariance function:

- $\mathbb{E}X_t^2 = \Gamma(t, t) = t$ for all $t \geq 0$; in particular, $X_0 = 0$ a.s.
- the increments $X_t - X_s$ follow a normal distribution with mean zero and variance $\mathbb{E}(X_t - X_s)^2 = \mathbb{E}X_t^2 - 2\mathbb{E}X_tX_s + \mathbb{E}X_s^2 = t - 2s \wedge t + s = |t - s|$; hence, the increments have a stationary distribution.
- any two increments are uncorrelated: for any $0 \leq u < v \leq s < t$ we have $\mathbb{E}[(X_v - X_u)(X_t - X_s)] = v - v - u + v = 0$; likewise, any finite number of increments are uncorrelated, hence independent.
- for any $t_1 \leq t_2 \leq \dots \leq t_n \leq s \leq t$,

$$\begin{aligned} \mathbb{P}(X_{t_1} \in A_1, \dots, X_{t_n} \in A_n, X_t - X_s \in A) &= \mathbb{P}((X_{t_1}, X_{t_2} - X_{t_1}, \dots, X_{t_n} - X_{t_{n-1}}) \in C, X_t - X_s \in A) \\ &= \mathbb{P}((X_{t_1}, X_{t_2} - X_{t_1}, \dots, X_{t_n} - X_{t_{n-1}}) \in C) \mathbb{P}(X_t - X_s \in A) \\ &= \mathbb{P}(X_{t_1} \in A_1, \dots, X_{t_n} \in A_n) \mathbb{P}(X_t - X_s \in A) \end{aligned}$$

where $C := \{(y_1, \dots, y_n) \in \mathbb{R}^n : y_1 \in A_1, y_1 + y_2 \in A_2, \dots, y_1 + \dots + y_n \in A_n\}$; hence

$$X_t - X_s \perp\!\!\!\perp X_{t_1}, \dots, X_{t_n}.$$

Thus, the process X satisfies the desired properties, except for the continuity of the sample paths. This can be guaranteed by the Kolmogorov-Centsov theorem.

Corollary 2.1.11 (Existence of Brownian motion) *Let X be a continuous Gaussian process with mean zero and covariance function $\Gamma(s, t) = s \wedge t$. Then a continuous modification B exists. In particular, a standard Brownian motion exists.*

PROOF: Since $X_t - X_s \sim N(0, t - s)$ there exists a constant C_n such that

$$\mathbb{E}|X_t - X_s|^{2n} = C_n |t - s|^n.$$

Thus, by Kolmogorov-Centsov, for each $T > 0$ there exists a continuous modification B^T of X on $[0, T]$. We need to show that a continuous modification exists on $[0, \infty)$. To this end, let

$$\Omega_T := \{\omega : B_t^T(\omega) = X_t(\omega) \text{ for all } t \in [0, T] \cap \mathbb{Q}\}; \quad \tilde{\Omega} := \bigcap_{T=1}^{\infty} \Omega^T.$$

Then $\mathbb{P}(\tilde{\Omega}) = 1$ and for all $\omega \in \tilde{\Omega}$ and any $T_1 \leq T_2 \in \mathbb{N}$,

$$B_t^{T_1}(\omega) = B_t^{T_2}(\omega), \quad t \in [0, T_1] \cap \mathbb{Q}$$

and by continuity of B^{T_1} and B^{T_2} on $[0, T_1]$ we get $B_t^{T_1}(\omega) = B_t^{T_2}(\omega)$ for all $t \in [0, T_1]$. Thus, we can define $B_t(\omega)$ as the respective common values for $\omega \in \tilde{\Omega}$ and $B_t(\omega) = 0$ for $\omega \in \tilde{\Omega}^c$. $\square$

The Kolmogorov-Centsov theorem shows that Brownian sample paths are even Hölder continuous with exponent $0 \leq \alpha < \frac{1}{2}$.

Proposition 2.1.12 $0 \leq \alpha < \frac{1}{2}$, almost every trajectory of a Brownian motion belongs to C^α .

Exercise 18 Show that the processes $(B_t^2 - t)_{t \geq 0}$ and $\left(e^{\alpha B_t - \frac{\alpha^2}{2}t}\right)_{t \geq 0}$ ($\alpha \in \mathbb{R}$) are martingales. In particular, a Brownian motion admits an exponential moment.

2.2 The Wiener-Lévy construction

The sample space for the Brownian motion constructed in the previous section was $\mathbb{R}^{[0, \infty)}$ the space of all real-valued functions on $[0, \infty)$. In this section, we provide a second construction of Brownian motion by Fourier expansion of the trajectories. In this case, the sample space will be a space Ω that is rich enough to carry a countable collection of independent, standard normal random variables.

To simplify the notation we restrict our construction to the time interval $[0, 1]$ and consider the space $L^2[0, 1]$ equipped with the usual inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$. We recall that $\{\phi^n\} \subset L^2$ is called a complete orthonormal system (CONS) if $\langle \phi^n, \phi^m \rangle = \delta_{n,m}$ and its span is dense in L^2 . In this case, any $f \in L^2$ admits the representation

$$f = \sum_{n=0}^{\infty} \langle f, \phi^n \rangle \phi^n,$$

that is, $f_m := \sum_{n=0}^m \langle f, \phi^n \rangle \phi^n \rightarrow f$ in L^2 . In particular, $\|f_m\|_2^2 = \sum_{n=0}^m \langle f, \phi^n \rangle^2 \rightarrow \|f\|_2^2 = \sum_{n=0}^{\infty} \langle f, \phi^n \rangle^2$. Moreover, the Parseval identity holds:

$$\begin{aligned} \langle f, g \rangle &= \frac{1}{4} (\|f + g\|^2 - \|f - g\|^2) \\ &= \frac{1}{4} \left(\sum_{n=0}^{\infty} \langle f + g, \phi^n \rangle^2 - \sum_{n=0}^{\infty} \langle f - g, \phi^n \rangle^2 \right) \\ &= \sum_{n=0}^{\infty} \langle f, \phi^n \rangle \langle g, \phi^n \rangle. \end{aligned}$$

We now show that one can use Fourier expansions of this type for the construction of Brownian motion. To this end, let $0 \leq s, t \leq 1$, $f = \mathbf{1}_{[0,t]}$ and $g = \mathbf{1}_{[0,s]}$. Then

$$s \wedge t = \langle f, g \rangle = \sum_{n=0}^{\infty} \langle f, \phi^n \rangle \langle g, \phi^n \rangle.$$

Let us now fix a sequence $\{Z_n\}$ of i.i.d. $N(0,1)$ distributed random variables and assume for the moment that the following definition makes sense:

$$X_t := \sum_{n=0}^{\infty} Z_n \langle \mathbf{1}_{[0,t]}, \phi^n \rangle = \sum_{n=0}^{\infty} Z_n \int_0^t \phi^n(x) dx.$$

Then, heuristically $\mathbb{E}X_t = 0$ and

$$\mathbb{E}[X_t X_s] = \sum_{n,m=0}^{\infty} \mathbb{E}[Z_n Z_m] \langle \mathbf{1}_{[0,t]}, \phi^n \rangle \langle \mathbf{1}_{[0,s]}, \phi^m \rangle = s \wedge t.$$

This suggests that X is a Brownian motion on $[0, 1]$.

We now make this construction rigorous. We first notice that we can choose *any* CONS for our construction. We use the *Haar basis*; it can be constructed by translations and scalings of the base function

$$H(t) = \begin{cases} 1, & 0 \leq t < \frac{1}{2} \\ -1 & \frac{1}{2} \leq t < 1 \\ 0 & \text{else} \end{cases}.$$

Putting $H_0(t) = \mathbf{1}_{[0,t]}$ and $H_1(t) = H(t)$, we can then define a sequence of functions $\{H_n\}$ via

$$H_n(t) = 2^{j/2} H(2^j t - k), \quad n = 2^j + k, \quad 0 \leq k < 2^j.$$

The function H_n is supported on the right-open interval $I_n = [k2^{-j}, (k+1)2^{-j})$ and has integral zero and norm one in the Hilbert space $L^2(\mathbb{R})$. We leave it as an exercise to show that this system forms an CONS of $L^2[0,1]$ (the set of indicator functions on dyadic intervals is dense; definition of integral through integrals of step functions).

The description of a Brownian motion X uses the functions $\int_0^t H_n(x) dx$ for which we now derive a convenient representation. Let us put

$$G(t) := 2 \int_0^t H(x) dx = \begin{cases} 2t, & 0 \leq t < \frac{1}{2} \\ 2(1-t) & \frac{1}{2} \leq t < 1 \\ 0 & \text{else} \end{cases}.$$

For $n \geq 0$, let $G_0(t) = t$, $G_1(t) = G(t)$, and

$$G_n(t) = G(2^j t - k), \quad n = 2^j + k, \quad 0 \leq k < 2^j.$$

Then, $0 \leq G_n \leq 1$ on $[0,1]$. Moreover,

$$\begin{aligned} \int_0^t H_n(x) dx &= \int_0^t 2^{j/2} H(2^j x - k) dx \\ &= \int_{-k}^{2^j t - k} H(s) 2^{j/2} 2^{-j} ds \quad (s = 2^j x - k) \\ &= \int_0^{2^j t - k} H(s) 2^{-j/2} ds \quad (H = 0 \text{ on } \mathbb{R}_-) \\ &= \lambda_n G_n(t), \quad \lambda_n := 2^{-\frac{j}{2}-1}. \end{aligned}$$

Remark 2.2.1 We notice that for $n = 2^j + k$ and $n' = 2^{j'} + k'$ if $k \neq k'$, then $G_n(t)G_{n'}(t) = 0$. This important result will be used repeatedly in what follows.

Before we state and prove the main result of this section, we establish the following useful estimate on large values of sequences of normal random variables.

Lemma 2.2.2 Let $\{Z_n\}$ be a sequence of i.i.d. $N(0,1)$ distributed random variables. Then there exists an a.s. finite random variable C such that almost surely, for all $n \geq 2$,

$$|Z_n| \leq C \sqrt{\log n}.$$

PROOF: For any $x \geq 1$,

$$\begin{aligned}\mathbb{P}(|Z_n| \geq x) &= 2 \int_x^\infty \frac{1}{\sqrt{2\pi}} \exp(-t^2/2) dt \\ &\leq \sqrt{\frac{2}{\pi}} \int_x^\infty t \exp(-t^2/2) dt \\ &= \sqrt{\frac{2}{\pi}} \left[-\exp(-t^2/2) \right]_x^\infty \\ &= \sqrt{\frac{2}{\pi}} \exp(-x^2/2).\end{aligned}$$

Hence, for all $\alpha > 1$ and $n \geq 2$,

$$\mathbb{P}(|Z_n| \geq \sqrt{2\alpha \log n}) \leq \sqrt{\frac{2}{\pi}} n^{-\alpha}.$$

In particular, $\sum_n \mathbb{P}(|Z_n| \geq \sqrt{2\alpha \log n}) < \infty$ and so it follows by the Borel-Cantelli lemma that

$$|Z_n| \leq \sqrt{2\alpha \log n}, \quad \text{for all } n \geq n(\omega).$$

In particular,

$$C(\omega) := \sup_{n \geq 2} \frac{|Z_n(\omega)|}{\sqrt{\log n}} < \infty \quad \text{a.s.}$$

□

Theorem 2.2.3 *Let $\{Z_n\}$ be a sequence of i.i.d. $N(0,1)$ distributed random variables. Then,*

$$X_t := \sum_{n=0}^{\infty} \lambda_n Z_n G_n(t), \quad t \in [0, 1]$$

converges uniformly on $[0, 1]$ almost surely and X is a Brownian motion.

PROOF: We first prove absolute uniform convergence of the series. This implies that X is a.s. continuous. Let $J, M \in \mathbb{N}$. For $M \geq 2^J$ and $t \in [0, 1]$ in view of the preceding lemma and using Remark 2.2.1 in the third step,

$$\begin{aligned}\sum_{n=M}^{\infty} \lambda_n |Z_n| G_n(t) &\leq C \sum_{n=M}^{\infty} \lambda_n \sqrt{\log n} G_n(t) \\ &\leq C \sum_{j=J}^{\infty} \sum_{k=0}^{2^j-1} 2^{-j/2-1} \sqrt{j+1} G_{2^j+k}(t) \quad (n \leq 2^{j+1}; \log n \leq \log(2)(j+1) \leq j+1) \\ &\leq C \sum_{j=J}^{\infty} 2^{-j/2-1} \sqrt{j+1} \quad (G \leq 1, \text{ only one of the } G\text{'s in the second sum is nonzero}) \\ &\rightarrow 0 \quad \text{as } J \rightarrow \infty, \text{ uniformly in } t \in [0, 1].\end{aligned}$$

Next, we prove that X has the correct covariance structure. To this end, we first notice that the above series forms a Cauchy sequence in L^2 . In fact, for any $m \leq n$, and $2^{j_0} \leq m$ using the same arguments as above (expanding the sum as a double sum, then using that only one of the G 's in the second sum is different from zero) it holds that

$$\mathbb{E} \left[\sum_{k=m+1}^n \lambda_k Z_k G_k(t) \right]^2 = \sum_{k=m+1}^n \lambda_k^2 G_k^2(t) \leq \sum_{j \geq j_0} 2^{-j-1} \rightarrow 0$$

for $j_0 \rightarrow \infty$. Since X_s exists as an L^2 limit,

$$\mathbb{E}[(X_t \pm X_s)^2] = \sum_{k=0}^{\infty} \lambda_k^2 (G_k(t) \pm G_k(s))^2$$

and by polarisation

$$\begin{aligned} \mathbb{E}[X_t X_s] &= \frac{1}{4} \mathbb{E}[(X_t + X_s)^2] - \frac{1}{4} \mathbb{E}[(X_t - X_s)^2] \\ &= \sum_{k=0}^{\infty} \lambda_k^2 G_k(t) G_k(s) \\ &= \sum_{k=0}^{\infty} \langle \mathbf{1}_{[0,t]}, H_k \rangle \langle \mathbf{1}_{[0,s]}, H_k \rangle \\ &= \langle \mathbf{1}_{[0,t]}, \mathbf{1}_{[0,s]} \rangle \quad (\text{Parseval identity}) \\ &= s \wedge t. \end{aligned}$$

It remains to prove that X is a centered Gaussian process. This is left as an exercise. $\square$

Exercise 19 *Prove that the process X is Gaussian with $\mu(t) = 0$ and $\text{cov}(X_s, X_t) = s \wedge t$.*

2.3 Weak convergence and the Wiener measure

So far we have established the existence of a probability measure $\mathbb{P}$ on $(\mathbb{R}^{[0,T]}, \mathcal{B}^{[0,T]})$ such that the canonical process has the distributional properties of a Brownian motion and that there exists a modification with a.s. continuous sample paths. It is thus natural to ask if we can work right away on the set

$$\Omega := C([0, \infty), \mathbb{R}) \subset \mathbb{R}^{[0, \infty)}$$

of all real-valued continuous functions on $[0, \infty)$. In other words, can we construct a measure on the set of continuous functions such that the canonical process

$$X_t : \Omega \rightarrow \mathbb{R}, \quad \omega \mapsto \omega(t), \quad t \geq 0$$

is a Brownian motion? To do this, we introduce on Ω the metric

$$d(f, g) := \sum_{n=1}^{\infty} \frac{1}{2^n} \max_{0 \leq t \leq n} (|f(t) - g(t)| \wedge 1) =: \sum_{n=1}^{\infty} \frac{1}{2^n} d_n(f, g).$$

Theorem 2.3.1 *The space (Ω, d) is a complete, separable metric space. The metric d induces the topology of uniform convergence on Ω .*

PROOF: Each d_n is a complete metric on $(C([0, n], \mathbb{R}), \|\cdot\|_{\infty})$. Let (f_N) be a Cauchy sequence in (Ω, d) . Then there exists a function g_n such that $d_n(f_N, g_n) \rightarrow 0$ as $N \rightarrow \infty$. For $x \leq n \leq m$ we have that $g_n(x) = g_m(x)$ and so there exists $g \in \Omega$ whose restriction to $[0, n]$ coincides with g_n , for each $n \in \mathbb{N}$. Then, $d(f_N, g) \rightarrow 0$. Hence Ω is complete. Moreover, by Weierstrass' theorem, the countable class of polynomials with rational coefficients is dense in $(C([0, n], \mathbb{R}), \|\cdot\|_{\infty})$, for each $n \in \mathbb{N}$ and is hence dense in (Ω, d) as well. $\square$

In view of the previous theorem, we can equip $C([0, \infty), \mathbb{R}) \subset \mathbb{R}^{[0, \infty)}$ with its Borel- σ -field $\mathcal{B}(C)$. The following exercise shows that $\mathcal{B}(C)$ is generated by the continuous functions as well as by the cylinder sets (hence projection mappings).

Exercise 20 (*) Let us consider the following σ -algebras on $C = C[0, 1]$:

- the Borel sigma-algebra $\mathcal{B}(C)$;
- the sigma-algebra $\sigma(C)$ generated by continuous functions $f : [0, 1] \rightarrow \mathbb{R}$, i.e.

$$\sigma(C) := \sigma \left(\bigcup_{f \in C} \{f^{-1}(B), B \in \mathcal{B}(\mathbb{R})\} \right);$$

- the sigma-algebra $\sigma(\mathcal{Z})$ generated by the system $\mathcal{Z}$ of all cylinder sets

$$Z = \{x \in C : x(t_j) \in A_j, j = 1, \dots, n\},$$

with $n \in \mathbb{N}, 0 \leq t_1 < t_2 < \dots < t_n \leq 1$ and $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$.

Show that

$$\sigma(C) = \mathcal{B}(C) = \sigma(\mathcal{Z}).$$

In particular, the mapping $X : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{B}(S))$, $\omega \mapsto X(\omega)$ is measurable if and only if for all $t_i \in [0, 1]$ the coordinate evaluation mapping $\omega \mapsto X_{t_i}(\omega)$ is measurable.

Hint. To show that $\mathcal{B}(S) \subset \sigma(\mathcal{Z})$ show first that every closed ε -ball is in $\sigma(\mathcal{Z})$ and then use separability of the metric space S to conclude. To show that $\mathcal{B}(S) \subset \sigma(C(S))$ let $A \subset S$ be closed on find a function $h \in C(S)$ such that $A = \{h = 0\}$.

Let B be a Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then there exists a measurable set $\bar{\Omega}$ of full measure such that $B(\bar{\omega}) \in C([0, \infty), \mathbb{R})$ for all $\bar{\omega} \in \bar{\Omega}$. Let $\mathcal{A} := \mathcal{F}|_{\bar{\Omega}}$ and $\bar{\mathbb{P}} := \mathbb{P}|_{\bar{\Omega}}$. Then $B : \bar{\Omega} \rightarrow C([0, \infty), \mathbb{R})$ is $(\bar{\mathcal{A}}, \mathcal{B}(C))$ -measurable,

$$\mathbb{P} := \bar{\mathbb{P}} \circ B^{-1}$$

is a measure on $C([0, \infty), \mathbb{R})$, and the canonical process X on $C([0, \infty), \mathbb{R})$ is a Brownian motion. Further details can be found in my Stochastic II lecture notes.

Definition 2.3.2 Let $\mathbb{P}$ be the probability measure on $\Omega = C([0, \infty), \mathbb{R})$ w.r.t. which the canonical process is a Brownian motion. Then $\mathbb{P}$ is called the Wiener measure. The triple $(\Omega, \mathcal{B}(C), \mathbb{P})$ is called the Wiener space. This is the canonical model when working with Brownian motion.

In what follows we are going to show that there exists a sequence of measures $\{\mathbb{P}_n\}$ on $(C, \mathcal{B}(C))$ - namely the distributions of a sequence of rescaled and linearly interpolated random walks - that converges to the Wiener measure. This provides a third construction of Brownian motion.

2.3.1 Prohorov's theorem and tightness of probability measures

This goal of this section is to establish a general condition that guarantees that a family of probability measures on a Polish space is weakly relatively sequentially compact, i.e. has a weakly convergent subsequence. We are particularly interested in families of probability measures on the space $C([0, 1], \mathbb{R})$. This result will be used to provide an alternative proof to Donsker's invariance principle.

We recall that a sequence of probability measures $(\mathbb{P}_n)_{n \in \mathbb{N}}$ on a metric space $(E, \mathcal{E})$ converges weakly to some probability measure $\mathbb{P}$ on $(E, \mathcal{E})$, written $\mathbb{P}_n \Rightarrow \mathbb{P}$, if

$$\int_E g(x) \mathbb{P}_n(dx) \rightarrow \int_E g(x) \mathbb{P}(dx) \quad \text{for all bounded and continuous function } g : E \rightarrow \mathbb{R}.$$

Definition 2.3.3 Let $(E, \mathcal{B}(E))$ be a metric space with Borel σ -field.

- A family of probability measures $\mathcal{P}$ on $(E, \mathcal{B}(E))$ is called
 - a) *relatively (sequentially) compact*, if every $(\mathbb{P}_n) \subset \mathcal{P}$ contains a subsequence, which converges weakly to some probability measure $\mathbb{Q}$ on $(E, \mathcal{B}(E))$, i.e. if there exists a subsequence such that

$$\int_E g(x) \mathbb{P}_{n_k}(dx) \longrightarrow \int_E g(x) \mathbb{P}(dx)$$

for all bounded and continuous $g : E \rightarrow \mathbb{R}$.

- b) *tight*, if for every $\varepsilon > 0$ there exists a compact set K such that $\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(K^c) < \varepsilon$.
- A family of random variables $\mathcal{X}$ taking values in $(E, \mathcal{B}(E))$ is called *tight*, if for every $\varepsilon > 0$ there exists a compact set $K \in E$ such that $\sup_{X \in \mathcal{X}} \mathbb{P}(X \in K^c) < \varepsilon$.

Theorem 2.3.4 (Prohorov) A family of probability measures $\mathcal{P}$ on a Polish space (E, d) is tight, if and only if it is weakly relatively sequentially compact.

The proof of Prohorov's theorem is involved. The fact that weakly relative sequentially compact implies tightness is relatively easy and established by the following lemma.

Lemma 2.3.5 If a family of probability measures $\mathcal{P}$ on a Polish space $(E, \mathcal{B}(E))$ is weakly relatively sequentially compact, then it is tight.

PROOF: Let $\{x_n\} \subset E$ be dense and for any $n \in \mathbb{N}$, let

$$A_{n,N} := \bigcup_{i=1}^N B_{1/n}(x_i).$$

Then $A_{n,N} \uparrow E$ as $N \rightarrow \infty$ for any $n \in \mathbb{N}$. We claim that

$$\delta := \sup_n \inf_N \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(A_{n,N}^c) = 0.$$

In fact, there exists $n \in \mathbb{N}$ such that for every $N \in \mathbb{N}$ there exists $\mathbb{P}_N \in \mathcal{P}$ such that $\mathbb{P}_N(A_{n,N}^c) \geq \delta/2$. Let $(\mathbb{P}_{N_k})$ converge weakly to $\mathbb{P}$. By the Portmanteau theorem

$$\mathbb{P}(A_{n,N}^c) \geq \liminf_k \mathbb{P}_{N_k}(A_{n,N}^c) \geq \liminf_k \mathbb{P}_{N_k}(A_{n,N_k}^c) \geq \delta/2.$$

At the same time $A_{n,N}^c \downarrow \emptyset$ so $\mathbb{P}(A_{n,N}^c) \rightarrow 0$, hence $\delta = 0$.

Let us now fix $\varepsilon > 0$. We have just shown that for any $n \in \mathbb{N}$ we can choose $N'_n \in \mathbb{N}$ such that $\mathbb{P}(A_{n,N'_n}^c) < \frac{\varepsilon}{2^n}$ for any $\mathbb{P} \in \mathcal{P}$. By construction, the set

$$A := \bigcap_{n \in \mathbb{N}} A_{n,N'_n}$$

is totally bounded, hence relatively compact. Tightness now follows from

$$\mathbb{P}((\bar{A})^c) \leq \mathbb{P}(A^c) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_{n, N'_n}^c) \leq \varepsilon, \quad \mathbb{P} \in \mathcal{P}.$$

□

The converse implication in Prohorov's theorem is much harder to prove. The proof for $E = \mathbb{R}$ is simple and given below. We are particularly interested in the case $E = C([0, T], \mathbb{R})$. This space is not locally compact hence we cannot apply the standard Riesz representation theorem (see below proof). But the space is Polish (complete, separable, metric space). For Polish spaces we first prove the result for compact spaces and then for general ones.

The case $E = \mathbb{R}$

If $E = \mathbb{R}$, then the proof of Prohorov's theorem is essentially a corollary of Helly's selection theorem.

Proposition 2.3.6 (Helly's selection theorem) *Let V be the class of all right-continuous, increasing and bounded function $F : \mathbb{R} \rightarrow \mathbb{R}$. Let $\{F_n\} \subset V$ be uniformly bounded. Then there exists $F \in V$ and a subsequence $\{F_{n_k}\}$ such that*

$$F_{n_k}(x) \rightarrow F(x) \quad \text{for all } x \in \mathbb{R} \text{ in which } F \text{ is continuous.}$$

PROOF: The proof is based on a diagonal argument. Since the set of rational numbers is countable, we may write $\mathbb{Q} = \{q_n\}$. The theorem of Bolzano-Weierstrass guarantees that the sequence $\{F_n(q_1)\}$ converges along a subsequence $\{n_k^1\}$. This sequence has a further subsequence $\{n_k^2\}$ such that $\{F_{n_k^2}(q_2)\}$ converges; etc. Iteratively, we obtain subsequences $\{n_k^1\} \supseteq \{n_k^2\} \supseteq \{n_k^3\} \supseteq \dots$ such that $\{F_{n_k^l}(q_l)\}$ converges for every $l \in \mathbb{N}$. Let $n_k := n_k^k$. Then $\{F_{n_k}(q)\}$ converges, for every $q \in \mathbb{Q}$. Let $\tilde{F}(q) := \lim_k F_{n_k}(q)$ and

$$F(x) := \inf\{\tilde{F}(q) : q \in \mathbb{Q}, q > x\}.$$

Since $\tilde{F}$ is increasing, F is right-continuous and increasing. If F is continuous in x , then for any $\varepsilon > 0$ we can choose, $q_1, q_2 \in \mathbb{Q}$ with $q_1 \leq x \leq q_2$ and

$$\tilde{F}(q_1) \geq F(x) - \varepsilon; \quad \tilde{F}(q_2) \leq F(x) + \varepsilon.$$

Thus, by construction,

$$\limsup_k F_{n_k}(x) \leq \lim_k F_{n_k}(q_2) = \tilde{F}(q_2) \leq F(x) + \varepsilon$$

and so $\limsup_k F_{n_k}(x) \leq F(x)$; likewise $\liminf_k F_{n_k}(x) \geq F(x)$. □

We emphasize that Helly's theorem states that the set V is sequentially compact; it does *not* state that the set of distribution functions is sequentially compact as we do not know the values of F at $\pm\infty$. This is precisely the reason why we need tightness of a family of probability measures to obtain the convergence of their distribution functions to a distribution function along a subsequence.

Corollary 2.3.7 *Any tight family of probability measures $\mathcal{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is weakly relatively sequentially compact.*

PROOF: Now, let $\mathcal{P}$ be a tight sequence of probability measures on $\mathbb{R}$ and $\{\mathbb{P}_n\} \subseteq \mathcal{P}$. Let F_n be the distribution function of $\mathbb{P}_n$ and let F be bounded, right-continuous, increasing function such that $F_{n_k}(x) \rightarrow F(x)$ for all $x \in \mathbb{R}$ in which F is continuous. We need to show that F is a distribution function, that is, that $F(-\infty) = 0$ and $F(+\infty) = 1$.

By the tightness, there exists for all $\varepsilon > 0$ a constant $K > 0$ such that $F_n(x) < \varepsilon$ for all $x < -K$ and all $n \in \mathbb{N}$. Let $x < -K$ be a point of continuity of F , e.g. a rational number. Then, $F_{n_k}(x) \rightarrow F(x)$ and hence

$$0 = \liminf_k F_{n_k}(-\infty) \geq \liminf_k F_{n_k}(x) - \varepsilon = F(x) - \varepsilon \geq F(-\infty) - \varepsilon \geq -\varepsilon.$$

Hence $F(-\infty) = 0$. Likewise, we obtain that $F(+\infty) \geq 1$. At the same time, by construction,

$$F(+\infty) \leq \liminf_n F_n(+\infty) = 1.$$

Thus, F is a distribution function and it follows from the Portemanteau theorem that $\mathbb{P}_n \rightarrow \mathbb{P}$. $\square$

Example 2.3.8 *This example illustrates that pointwise limits of sequences of distribution functions do not need to be distribution functions. In fact, let F be a distribution functions and let*

$$F_n(x) := F(x + n); \quad G_n(x) := F(x - n).$$

Then, $F_n \rightarrow \tilde{F} \equiv 1$ and $G_n \rightarrow \tilde{G} \equiv 0$ pointwise, which are obviously not distribution functions.

It is not difficult to show that the set V is a metric space under the Lévy metric

$$L(F, G) := \inf\{\varepsilon > 0 : F(x - \varepsilon) - \varepsilon \leq G(x) \leq F(x + \varepsilon) + \varepsilon \text{ for all } x \in \mathbb{R}\}.$$

Furthermore, if (F_n) is a sequence of distribution functions and F is also a distribution function with corresponding probability measures μ_n and μ , then $\mu_n \rightarrow \mu$ weakly if and only if $L(F_n, F) \rightarrow 0$.

Exercise 21 (*)

- Prove that L is a metric on the space of distribution functions.
- Let (F_n) and F be distribution functions with corresponding probability measures μ_n and μ . Prove $\mu_n \rightarrow \mu$ weakly if and only if $L(F_n, F) \rightarrow 0$.

In particular, the set $\mathcal{P}(\mathbb{R})$ of probability measures on $\mathbb{R}$ is metrizable and complete by Helly's theorem. The next subsection establishes a similar result for probability measures on Polish spaces.

The case of Polish spaces.

In what follows we prove Prohorov's theorem for general Polish spaces (E, d) . We start with the following result that shows that every probability measure on a Polish space is tight.

Proposition 2.3.9 (Tightness on Polish spaces) *Let (E, d) be a Polish space. Then every Borel probability measure μ on E is tight, i.e. for every $\varepsilon > 0$ there exists a compact set $K \subset E$ such that*

$$\mu(K) \geq 1 - \varepsilon.$$

PROOF: Since E is Polish, it is separable. Let $(x_k)_{k \in \mathbb{N}}$ be a dense sequence in E . For each $m \in \mathbb{N}$ define the open sets

$$U_m := \bigcup_{k=1}^m B(x_k, 1/m).$$

Then $(U_m)_{m \in \mathbb{N}}$ is an increasing sequence of open sets and

$$\bigcup_{m=1}^{\infty} U_m = E.$$

By continuity from below, $\mu(U_m) \uparrow 1$. Hence there exists m such that $\mu(U_m) \geq 1 - \varepsilon$. The set

$$K := \overline{U_m}.$$

is closed and $\mu(K) \geq \mu(U_m) \geq 1 - \varepsilon$. Since K is totally bounded by construction, it is compact since E is complete. □

We now recall that continuous functions on compact metric spaces can be approximated by Lipschitz continuous functions. For instance, if K is a compact subset of some metric space X and $f \in C(K)$, then the functions

$$f_n(x) := \inf_{y \in X} (f(y) + n d(x, y))$$

are Lipschitz with constant n and converge uniformly to f on K . This allows us to prove the following two results.

Proposition 2.3.10 *Let (K, d) be a compact metric space. Then $C(K)$ is complete and separable.*

PROOF: The proof of completeness is standard. Separability follows from the above approximation result as follows. Since K is compact metric, it is separable. Let

$$D = \{x_1, x_2, x_3, \dots\} \subset K$$

be a dense subset. For any $N \in \mathbb{N}$ and any rational numbers $q_1, \dots, q_N \in \mathbb{Q}$ satisfying

$$|q_i - q_j| \leq L d(x_i, x_j),$$

we define the function

$$g(x) := \min_{1 \leq i \leq N} (q_i + L d(x, x_i)).$$

Each such function is Lipschitz continuous with constant L . Let $\mathcal{A}_L$ denote the set of all these. Then $\mathcal{A}_L$ is countable. Since any continuous function on K can be uniformly approximated by Lipschitz functions it is enough to prove that $\mathcal{A}_L$ is dense in the class of Lipschitz continuous functions with constant L .

To this end, let $f : K \rightarrow \mathbb{R}$ be Lipschitz with constant L and let $\varepsilon > 0$. Choose $\delta := \varepsilon/(3L)$. Since D is dense and K is compact, we can find N such that $\{x_1, \dots, x_N\}$ is a δ -net, i.e., for every $x \in K$ there exists i with

$$d(x, x_i) < \delta.$$

¹We recall that subsets of complete metric spaces are compact if and only if they are closed and totally bounded.

We now choose $q_i \in \mathbb{Q}$ such that

$$|q_i - f(x_i)| < \varepsilon/3 \quad \text{for } i = 1, \dots, N$$

and define

$$g(x) := \min_{1 \leq i \leq N} (q_i + L d(x, x_i)).$$

We claim that $\|f - g\|_\infty < \varepsilon$. To prove an upper bound let $x \in K$ and choose i with $d(x, x_i) < \delta$. Then

$$g(x) \leq q_i + L d(x, x_i) \leq f(x_i) + \frac{\varepsilon}{3} + L\delta \leq f(x) + \frac{2\varepsilon}{3}$$

where the second inequality follows from the Lipschitz continuity of f and the definition of $\delta > 0$. To establish a lower bound, we note that for any i ,

$$q_i \geq f(x_i) - \frac{\varepsilon}{3}.$$

Thus, by Lipschitz continuity of f ,

$$q_i + L d(x, x_i) \geq f(x_i) - \frac{\varepsilon}{3} + L d(x, x_i) \geq f(x) - \frac{\varepsilon}{3}.$$

Taking the minimum over i yields

$$g(x) \geq f(x) - \frac{\varepsilon}{3}.$$

□

Proposition 2.3.11 (Bounded Lipschitz functions are convergence-determining) *Let (E, d) be a Polish space. Let μ_n, μ be probability measures on E . Then $\mu_n \Rightarrow \mu$ if*

$$\int f d\mu_n \rightarrow \int f d\mu \quad \text{for all bounded Lipschitz functions } f.$$

PROOF: Let $f \in C_b(E)$, fix $\varepsilon > 0$ and let $M := \|f\|_\infty$. By the tightness of μ , there exists a compact set $K \subset E$ such that

$$\mu(K) \geq 1 - \varepsilon.$$

The function

$$\varphi(x) := \max(0, 1 - d(x, K)).$$

is Lipschitz continuous, $0 \leq \varphi \leq 1$, and $\varphi = 1$ on K . Hence, the function $g := f\varphi$ has compact support, belongs to $C_b(E)$, $g = f$ on K , and

$$|f - g| \leq 2M \mathbf{1}_{K^c}.$$

As a result,

$$\left| \int f d\mu - \int g d\mu \right| \leq 2M \mu(K^c) \leq 2M\varepsilon \quad \text{and} \quad \left| \int f d\mu_n - \int g d\mu_n \right| \leq 2M \mu_n(K^c).$$

Since g has compact support we may w.l.o.g. assume that it is Lipschitz continuous. We now write

$$\left| \int f d\mu_n - \int f d\mu \right| \leq A_n + B_n + C_n,$$

where

$$A_n := \left| \int (f - g) d\mu_n \right|, \quad B_n := \left| \int g d\mu_n - \int g d\mu \right|, \quad C_n := \left| \int (g - f) d\mu \right|.$$

We know that $C_n \leq 2M\varepsilon$ that $B_n \rightarrow 0$, due to the Lipschitz continuity of g and that $A_n \leq 2M\mu_n(K^c)$. Hence it remains to control $\mu_n(K^c)$. For this, define

$$\psi_m(x) := \min(1, m d(x, K)).$$

Then ψ_m is bounded Lipschitz and $\psi_m \uparrow \mathbf{1}_{K^c}$ pointwise. Thus for each m large enough,

$$\int \psi_m d\mu_n \rightarrow \int \psi_m d\mu \leq \mu(K^c) + \varepsilon.$$

Thus,

$$\limsup_n \lim_m \int \psi_m d\mu_n = \limsup_{n \rightarrow \infty} \mu_n(K^c) \leq \mu(K^c) + \varepsilon.$$

□

The next result shows that weak convergence of probability measures on Polish spaces is metrizable (cf. the case $E = \mathbb{R}$ where a similar result was obtained).

Theorem 2.3.12 *Let (E, d) be a Polish space and let $\mathcal{P}(E)$ be the space of probability measures on E . Let*

$$d_{BL}(\mu, \nu) := \sup_{f \in \text{BL}_1(E)} \left| \int f d\mu - \int f d\nu \right|,$$

where $\text{BL}_1(E)$ denotes the set of all functions $f : E \rightarrow \mathbb{R}$ such that $\|f\|_\infty \leq 1$ and $\text{Lip}(f) \leq 1$. Then d_{BL} metrizes the weak convergence of probability measures.

PROOF: The preceding result shows that $\mu_n \Rightarrow \mu$ if $d_{BL}(\mu_n, \mu) \rightarrow 0$. Conversely, suppose that $\mu_n \Rightarrow \mu$ and fix $\varepsilon > 0$. By the easy implication of Prohorov's theorem our probability measures are tight: there exists a compact set $K \subset E$ such that

$$\mu_n(K) \geq 1 - \varepsilon, \quad \mu(K) \geq 1 - \varepsilon \quad \text{for all sufficiently large } n.$$

Let $f \in \text{BL}_1(E)$. Then

$$\int f d\mu_n - \int f d\mu = \int_K f d\mu_n - \int_K f d\mu + R_n \quad \text{where} \quad |R_n| \leq 2\varepsilon.$$

Since the family $\text{BL}_1(E)$ is equicontinuous and uniformly bounded, by Arzelà–Ascoli and Proposition 2.3.10 the restriction $\text{BL}_1(E)|_K$ is totally bounded in the sup norm. Hence there exist $f_1, \dots, f_m \in \text{BL}_1(E)$ such that for every $f \in \text{BL}_1(E)$ there exists i with

$$\sup_{x \in K} |f(x) - f_i(x)| < \varepsilon.$$

Thus,

$$\left| \int_K f d\mu_n - \int_K f d\mu \right| \leq \left| \int_K f_i d\mu_n - \int_K f_i d\mu \right| + 2\varepsilon.$$

Taking supremum over $f \in BL_1(E)$ gives

$$d_{BL}(\mu_n, \mu) \leq \max_{1 \leq i \leq m} \left| \int f_i d\mu_n - \int f_i d\mu \right| + 4\varepsilon.$$

Since each $f_i \in C_b(E)$ and $\mu_n \Rightarrow \mu$, we have

$$\int f_i d\mu_n \rightarrow \int f_i d\mu, \quad i = 1, \dots, m.$$

Hence the maximum term tends to zero, and therefore

$$\limsup_{n \rightarrow \infty} d_{BL}(\mu_n, \mu) \leq 4\varepsilon.$$

□

In a first step we now proof the complicated implication of Prohorov's theorem for probability measures on compact sets. The proof uses the Riesz representation theorem that we state without proof.

Theorem 2.3.13 (Riesz–Markov–Kakutani Representation Theorem) *Let X be a compact Hausdorff space and*

$$L : C(X) \rightarrow \mathbb{R}$$

be a positive linear functional, i.e. $L(f) \geq 0$ whenever $f \geq 0$. Then there exists a unique regular Borel measure μ on X such that

$$L(f) = \int_X f d\mu \quad \text{for all } f \in C(X).$$

Moreover,

$$\|L\| = \mu(X).$$

In particular, if $L(1) = 1$, then μ is a probability measure.

Proposition 2.3.14 (Sequential compactness on compact metric spaces) *Let K be a compact metric space. Then every sequence of probability measures $(\mu_n) \subset \mathcal{P}(K)$ admits a subsequence (μ_{n_k}) and a probability measure $\mu \in \mathcal{P}(K)$ such that*

$$d_{BL}(\mu_{n_k}, \mu) \rightarrow 0.$$

Equivalently, $\mathcal{P}(K)$ is weakly sequentially compact.

PROOF: Since K is compact metric, it follows from Proposition [2.3.10](#) that the space $C(K)$ is separable in the uniform norm. As a subset of the metric space $C(K)$ the set $BL_1(K)$ is also separable; we fix a countable dense family

$$(f_j)_{j \in \mathbb{N}} \subset BL_1(K).$$

For each fixed j , the sequence $\int f_j d\mu_n$ lies in $[-1, 1]$, hence admits a convergent subsequence. By a standard diagonal argument we find a subsequence (μ_{n_k}) such that for every $j \in \mathbb{N}$,

$$\int_K f_j d\mu_{n_k} \longrightarrow \ell_j \in \mathbb{R}.$$

Next, we show that the extracted subsequence (μ_{n_k}) is Cauchy in d_{BL} . Since the set $BL_1(K)$ is bounded and equicontinuous, it follows from the Arzelà–Ascoli theorem that it is compact, hence

totally bounded in $\|\cdot\|_\infty$. Hence, for any $\varepsilon > 0$ there exists a finite set $F_\varepsilon = \{f_1, \dots, f_J\} \subset BL_1(K)$ such that for every $f \in BL_1(K)$, there exists $j \in \{1, \dots, J\}$ with

$$\|f - f_j\|_\infty \leq \varepsilon.$$

Thus, for any k, m ,

$$\left| \int f d(\mu_{n_k} - \mu_{n_m}) \right| \leq \left| \int f_j d(\mu_{n_k} - \mu_{n_m}) \right| + 2\varepsilon.$$

Since

$$\max_{1 \leq j \leq J} \left| \int f_j d(\mu_{n_k} - \mu_{n_m}) \right| \rightarrow 0.$$

and $\varepsilon > 0$ was arbitrary we conclude that

$$d_{BL}(\mu_{n_k}, \mu_{n_m}) \rightarrow 0.$$

We now identify the limit via Riesz–Markov². For this, we define a linear functional on $BL_1(K)$ by

$$L(f) := \lim_{k \rightarrow \infty} \int f d\mu_{n_k}.$$

The functional is well-defined on $BL_1(K)$, linear, and satisfies:

$$|L(f)| \leq \|f\|_\infty, \quad L(f) \geq 0 \text{ if } f \geq 0, \quad L(1) = 1.$$

Since $BL_1(K)$ is dense in $C(K)$, the functional extends uniquely to a positive linear functional L on $C(K)$. Since K is compact Hausdorff, the Riesz–Markov–Kakutani representation theorem yields a unique probability measure $\mu \in \mathcal{P}(K)$ such that

$$L(f) = \int_K f d\mu, \quad f \in C(K).$$

In particular, for all $f \in BL_1(K)$,

$$\int f d\mu_{n_k} \rightarrow \int f d\mu.$$

Taking supremum over $\|f\|_{BL_1} \leq 1$, using again the total boundedness for $BL_1(K)$,

$$d_{BL}(\mu_{n_k}, \mu) = \sup_{\|f\|_{BL_1} \leq 1} \left| \int f d\mu_{n_k} - \int f d\mu \right| \rightarrow 0.$$

□

We are now ready to prove Prokhorov’s theorem on Polish spaces. The key difference is that for general Polish spaces we cannot apply the standard Riesz–Markov–Kakutani theorem which only works in locally compact topological spaces.

Theorem 2.3.15 *Let (E, d) be a Polish space and let $(\mu_n) \subset \mathcal{P}(E)$ be tight. Then there exists a subsequence (μ_{n_k}) and a probability measure $\mu \in \mathcal{P}(E)$ such that*

$$d_{BL}(\mu_{n_k}, \mu) \rightarrow 0.$$

²For this, the compactness (or at least local compactness) of K is required; compactness is not needed for the Arzelà–Ascoli theorem.

PROOF: Since (μ_n) is tight, for each $m \in \mathbb{N}$, there exists a compact set $K_m \subset E$ such that

$$\mu_n(K_m) \geq 1 - 2^{-m} \quad \text{for all } n \in \mathbb{N}.$$

We may w.l.o.g. assume that $K_m \subset K_{m+1}$. Fix m we consider the restricted measures (these are NOT probability measures)

$$\mu_n^{(m)} := \mu_n|_{K_m}.$$

Although the $\mu_n^{(m)}$ are not probability measures we can still use the same arguments as in the proof of the previous proposition to show that, for each $m \in \mathbb{N}$, the sequence $(\mu_n^{(m)})$ admits a subsequence converging in d_{BL} on K_m . Again by a standard diagonalization argument there exists a subsequence (μ_{n_k}) such that for every $m \in \mathbb{N}$, the restriction $\mu_{n_k}|_{K_m}$ converges in d_{BL} on K_m . Thus for each fixed $m \in \mathbb{N}$, there exists a finite Borel measure $\mu^{(m)}$ on K_m such that

$$d_{BL}(\mu_{n_k}|_{K_m}, \mu^{(m)}) \rightarrow 0.$$

Since $K_{m_1} \subset K_{m_2}$ whenever $m_1 < m_2$ and both limits agree on K_{m_1} , the family $(\mu^{(m)})$ is consistent.

To construct the global limit for $f \in BL(E)$ we set

$$L_m(f) := \int_{K_m} f d\mu^{(m)}$$

and claim that the sequence $(L_m(f))$ is Cauchy. Indeed, let $m < n$. Then, by consistency

$$L_n(f) = \int_{K_n} f d\mu^{(n)} = \int_{K_m} f d\mu^{(n)} + \int_{K_n \setminus K_m} f d\mu^{(n)} = L_m(f) + \int_{K_n \setminus K_m} f d\mu^{(n)}$$

and since $|f| \leq 1$,

$$\begin{aligned} |L_n(f) - L_m(f)| &\leq \mu^{(n)}(K_n \setminus K_m) \\ &= \lim_{k \rightarrow \infty} \mu_{n_k}(K_n \setminus K_m) \\ &\leq \limsup_{k \rightarrow \infty} \mu_{n_k}(K_m^c) \\ &\leq 2^{-m}. \end{aligned}$$

Since $(L_m(f))$ is Cauchy we can define the limit

$$L(f) := \lim_{m \rightarrow \infty} L_m(f).$$

Then L is linear, non-negative, normalized, and continuous under bounded convergence on compact sets. By the Riesz–Markov representation theorem for Radon measures on Polish spaces in the form of Proposition [2.3.16](#) below,³ there exists a unique probability measure μ such that

$$L(f) = \int_X f d\mu \quad \text{for all } f \in BL(E).$$

It remains to prove the convergence of the integrals. For this, we fix $f \in BL(E)$. For any $m \in \mathbb{N}$,

$$\int f d\mu_{n_k} = \int_{K_m} f d\mu_{n_k} + \int_{K_m^c} f d\mu_{n_k}.$$

³We emphasize that we assume that E but not necessarily locally compact. Hence we cannot use the standard Riesz representation theorem.

Since $|f| \leq 1$,

$$\left| \int_{K_m^c} f d\mu_{n_k} \right| \leq \mu_{n_k}(K_m^c) \leq 2^{-m}.$$

Also:

$$\int_{K_m} f d\mu_{n_k} \rightarrow \int_{K_m} f d\mu^{(m)} = L_m(f).$$

Thus:

$$\limsup_{k \rightarrow \infty} \left| \int f d\mu_{n_k} - L(f) \right| \leq 2^{-m}.$$

Letting $m \rightarrow \infty$, we obtain:

$$\int f d\mu_{n_k} \rightarrow \int f d\mu.$$

□

Proposition 2.3.16 *Let X be a Polish space and let $(\mu_n)_{n \in \mathbb{N}}$ be a sequence of probability measures on X . Assume:*

1. **Tightness:** *For every $\varepsilon > 0$, there exists a compact set $K \subset X$ such that*

$$\sup_n \mu_n(X \setminus K) < \varepsilon.$$

2. **Pointwise convergence on $C_b(X)$:** *There exists a bounded, positive, linear functional $T : C_b(X) \rightarrow \mathbb{R}$ such that*

$$\int_X f d\mu_n =: T_n(f) \rightarrow T(f) \quad \text{for all } f \in C_b(X).$$

Then there exists a unique probability measure μ on X such that

$$T(f) = \int_X f d\mu \quad \text{for all } f \in C_b(X).$$

PROOF: We first construct the candidate probability measure μ . For a compact set $K \subset X$, we set

$$\mu(K) := \inf \left\{ T(f) : f \in C_b(X), 0 \leq f \leq 1, f|_K = 1 \right\}.$$

This set-function is well-defined by Urysohn's lemma.⁴ It is monotone, finitely sub-additive and $\mu(\emptyset) = 0$. Moreover, $\mu_n(K) \rightarrow \mu(K)$. In fact, if $f \geq 1_K$, then

$$\mu_n(K) \leq \int f d\mu_n.$$

⁴If X is topological space where any two disjoint closed sets can be separated by disjoint open neighborhoods. Let $A, B \subset X$ be disjoint closed sets. Then there exists a continuous function

$$f : X \rightarrow [0, 1]$$

such that

$$f(x) = 0 \quad \text{for all } x \in A, \quad f(x) = 1 \quad \text{for all } x \in B.$$

Taking first the limsup over $n \in \mathbb{N}$ and then the infimum over all such f , we see that

$$\limsup_n \mu_n(K) \leq \mu(K).$$

To obtain the converse inequality and hence the desired convergence we fix $\varepsilon > 0$ and choose $f \in C_b(X)$ such that

$$0 \leq f \leq 1, \quad f|_K = 1, \quad T(f) \geq \mu(K) + \varepsilon.$$

Then

$$\mu_n(K) \geq \int f d\mu_n - \mu_n(K^c).$$

By tightness, the tail term is uniformly small. Hence

$$\liminf_n \mu_n(K) \geq T(f) \geq \mu(K) + \varepsilon.$$

We can extend μ to a probability measure on the Borel sets in the usual way⁵

To show that μ is the desired measure, we first show that it is enough to identify T on the set $C_c(X)$. In fact, let $f \in C_b(X)$ and $\varepsilon > 0$. By tightness, we can choose a compact set $K \subset X$ such that

$$\sup_n \mu_n(X \setminus K) < \varepsilon$$

and a function $\varphi \in C_c(X)$ such that $0 \leq \varphi \leq 1$, $\varphi|_K = 1$. Set $g := f\varphi \in C_c(X)$. Then

$$|T_n(f) - T_n(g)| \leq \|f\|_\infty \varepsilon \quad \text{and} \quad |T(f) - \lim_{n \rightarrow \infty} T_n(g)| \leq \|f\|_\infty \varepsilon.$$

To identify T on $C_c(X)$, let $g \in C_c(X)$, and let $K = \text{supp}(g)$, which is compact. Then

$$T_n(g) = \int_X g d\mu_n = \int_K g d\mu_n.$$

Since $\mu_n(K) \rightarrow \mu(K)$, and by standard approximation of continuous functions on a fixed compact set (using linear combinations of indicators of small sets and uniform continuity of g), it follows that

$$\int_K g d\mu_n \rightarrow \int_K g d\mu.$$

Hence,

$$T(g) = \int_X g d\mu \quad \text{for } g \in C_c(X).$$

For general continuous function $f \in C_b(X)$ we use again tightness to approximate f by a continuous function with compact support to show that $T(f) = \int f d\mu$. Finally, two probability measures agree if they agree on $C_b(X)$. $\square$

⁵For open sets $U \subset X$ we set $\mu(U) := \sup \{ \mu(K) : K \subset U, K \text{ compact} \}$ and for arbitrary sets A we set $\mu^*(A) := \inf \{ \mu(U) : A \subset U, U \text{ open} \}$. Then μ^* is an outer measure which extends uniquely to a measure on the Borel- σ -field.

Weak uniqueness of accumulation points

In any metric space a sequence converges if and only if any subsequence contains a further subsequence that converges to the same limit. The following lemma establishes a corresponding result for probability measures on general Borel spaces.

Lemma 2.3.17 *Let $(\mathbb{P}_n)_{n \in \mathbb{N}}$ and $\mathbb{P}$ be probability measures on a metric Borel space $(E, \mathcal{B}(E))$. Then $\mathbb{P}_n \Rightarrow \mathbb{P}$ if and only if every subsequence of $(\mathbb{P}_n)_{n \in \mathbb{N}}$ contains a further subsequence that converges weakly to $\mathbb{P}$.*

PROOF: The necessity is obvious. For sufficiency, suppose that $\mathbb{P}_n \Rightarrow \mathbb{P}$ does not hold. Then there exists a subsequence $(\mathbb{P}_{n_j}) \subset (\mathbb{P}_n)$, a bounded continuous function f and some $\varepsilon > 0$ such that

$$\left| \int_E f(x) \mathbb{P}_{n_j}(dx) - \int_E f(x) \mathbb{P}(dx) \right| > \varepsilon$$

for all $j \in \mathbb{N}$. Hence, no further subsequence of $\{\mathbb{P}_{n_j}\}_{j \in \mathbb{N}}$ can converge weakly to $\mathbb{P}$. $\square$

Corollary 2.3.18 *If $(\mathbb{P}_n)_{n \in \mathbb{N}}$ is a tight sequence of probability measures on a Polish space $(E, \mathcal{B}(E))$ and if each convergent subsequence converges to $\mathbb{P}$, then the entire sequence converges weakly to $\mathbb{P}$.*

PROOF: By Prokhorov's theorem, $(\mathbb{P}_n)_{n \in \mathbb{N}}$ is relatively sequentially compact, i.e. every subsequence contains a convergent further subsequence, which by assumption converges to $\mathbb{P}$. The result now follows from Lemma 2.3.17. $\square$

2.3.2 Weak convergence of probability measures on $C([0, 1], \mathbb{R})$

We are now going to prove convergence results for sequences of probability measures on $C([0, 1], \mathbb{R})$. Since $C([0, 1], \mathbb{R})$ equipped with the supremum-norm is a Polish space, Prokhorov's theorem is applicable in this setting. In particular, the weak convergence $\mathbb{P}_n \Rightarrow \mathbb{P}$ of a sequence of probability measures on $C([0, 1], \mathbb{R})$ implies the tightness of the sequence. It also implies convergence of the finite-dimensional distributions. In fact, for any finite set $J \subset [0, 1]$ the projection mapping

$$\pi_J : C([0, 1], \mathbb{R}) \rightarrow \mathbb{R}^{|J|}, \quad x \mapsto (x(t))_{t \in J}$$

is continuous. Hence, for any bounded continuous $g : \mathbb{R}^{|J|} \rightarrow \mathbb{R}$ it holds that

$$\int g(y) (\mathbb{P}_n \circ \pi_J^{-1})(dy) = \int g(\pi_J(x)) \mathbb{P}_n(dx) \xrightarrow{k} \int g(\pi_J(x)) \mathbb{P}(dx) = \int g(y) (\mathbb{P} \circ \pi_J^{-1})(dy). \quad (2.4)$$

The following result prove the converse implication. Hence a sequence a sequence of probability measures on $C([0, 1], \mathbb{R})$ converges weakly if and only if it is tight and the if the finite-dimensional distributions converge.

Theorem 2.3.19 *Let $(\mathbb{P}_n)_{n \in \mathbb{N}}$ and $\mathbb{P}$ be probability measures on $C([0, 1]; \mathbb{R})$. If the finite-dimensional distributions of the $\mathbb{P}_n$ converges weakly to those of $\mathbb{P}$ and if $(\mathbb{P}_n)_{n \in \mathbb{N}}$ is tight, then $\mathbb{P}_n \Rightarrow \mathbb{P}$.*

PROOF: If the sequence $(\mathbb{P}_n)$ is tight, then it is relatively sequentially compact. To prove the weak convergence, we would like to apply Corollary 2.3.18, for which we have to prove that all weakly convergent subsequences converge to $\mathbb{P}$.

To this end, let $(\mathbb{P}_{n_k})$ be a weakly convergent subsequence so that $\mathbb{P}_{n_k} \Rightarrow \mathbb{Q}$ for some probability measure $\mathbb{Q}$ on $C([0, 1]; \mathbb{R})$. We need to show that $\mathbb{P} = \mathbb{Q}$. By assumption,

$$\mathbb{P}_n \circ \pi_J^{-1} \Rightarrow \mathbb{P} \circ \pi_J^{-1} \quad \text{for all } J \subseteq [0, 1] \text{ finite.}$$

Moreover, it follows from (2.4) that $\mathbb{P}_{n_k} \circ \pi_J^{-1} \Rightarrow \mathbb{Q} \circ \pi_J^{-1}$. Hence

$$\mathbb{P} \circ \pi_J^{-1} = \mathbb{Q} \circ \pi_J^{-1} \quad \text{for all finite subsets } J \subseteq [0, 1].$$

Since a probability distribution on $C([0, 1]; \mathbb{R})$ is completely characterized by its finite-dimensional distributions (by Kolmogorov's consistency theorem), we conclude that $\mathbb{P} = \mathbb{Q}$. □

The following counterexample shows that the convergence of the finite-dimensional distributions of the $(\mathbb{P}_n)$ is in general not sufficient to have $\mathbb{P}_n \Rightarrow \mathbb{P}$.

Example 2.3.20 For every $n \in \mathbb{N}$, let us define a function $z_n : [0, 1] \rightarrow \mathbb{R}$ by

$$z_n(t) := nt \cdot \mathbb{1}_{[0, \frac{1}{n}]}(t) + (2 - nt) \mathbb{1}_{(\frac{1}{n}, \frac{2}{n}]}(t), \quad t \in [0, 1].$$

Then $\|z_n\|_\infty = 1$ for all $n \in \mathbb{N}$ and $z_n(t) \rightarrow 0$ for all $t \in [0, 1]$. Let

$$\mathbb{P}_n := \delta_{z_n}, \quad n \in \mathbb{N} \quad \text{and} \quad \mathbb{P} := \delta_0.$$

To show that the sequence $(\mathbb{P}_n)$ does not converge weakly to $\mathbb{P}$ we introduce the closed set $B := \{z \in C([0, 1]) : \|z\|_\infty = 1\}$. Then $\mathbb{P}(B) = 0$ and by the Portemanteau theorem weak convergence $\mathbb{P}_n \Rightarrow \mathbb{P}$ would give us that

$$1 = \limsup_n \mathbb{P}_n(B) \geq \mathbb{P}(B),$$

which does not hold. To show that we nonetheless have convergence of the finite dimensional distributions, let $0 = t_0 < t_1 < \dots < t_m \leq 1$. Then for all $n \in \mathbb{N}$ with $\frac{2}{n} < t_1$ we have

$$\pi_{\{t_0, \dots, t_m\}}(z_n) = (0, \dots, 0)$$

and so,

$$\mathbb{P}_n \circ \pi_{\{t_0, \dots, t_m\}}^{-1}(A) = \mathbb{P} \circ \pi_{\{t_0, \dots, t_m\}}^{-1}(A) \quad \text{for all } A \in \mathcal{B}(\mathbb{R}^{|J|}), \quad n > \frac{2}{t_1}.$$

Thus, $\mathbb{P}_n \circ \pi_J^{-1}(A) \rightarrow \mathbb{P} \circ \pi_J^{-1}(A)$ for all $J \subset [0, 1]$ finite and $A \in \mathcal{B}(\mathbb{R}^{|J|})$. That is, all finite-dimensional distributions converge.

In what follows we convert the classical Arzelà-Ascoli compactness criterion into a condition for the tightness of probability measures on $C(K, S)$ where (K, d) and (S, ϱ) are metric spaces. To this end, we recall that the *modulus of continuity* of a function $x \in C(K, S)$ is given by

$$w(x, h) := \sup\{\varrho(x_t, x_s) : d(t, s) \leq h\}, \quad h > 0.$$

The function w is continuous for fixed $h > 0$ and hence a measurable function of x . In what follows we need the following version of the Arzelà-Ascoli theorem.

Theorem 2.3.21 (Arzelà, Ascoli) *Let K and S be metric spaces where K is compact and S is complete, and let $D \subseteq K$ be dense. Then a set $A \subseteq C(K, S)$ is relatively compact if and only if $\{x_t : x \in A\}$ is relatively compact in S for any $t \in D$ and*

$$\limsup_{h \rightarrow 0} \sup_{x \in A} w(x, h) = 0.$$

In this case, even the set $\bigcup_{t \in K} \{x_t : x \in A\}$ is relatively compact in S .

We are now ready to prove the following tightness condition in $C(K, S)$, due to Prohorov.

Theorem 2.3.22 (Tightness in $C(K, S)$) *Let K and S be metric spaces where K is compact and S is separable and complete. Let $\{X_n\}$ be a sequence of random elements in $C(K, S)$. Then $X_n \xrightarrow{d} X$ if and only if the finite-dimensional distributions converge and*

$$\limsup_{h \rightarrow 0} \sup_n \mathbb{E}[w(X_n, h) \wedge 1] = 0. \quad (2.5)$$

PROOF: Since $C(K, S)$ is separable and complete, Prohorov's theorem shows that tightness and relative compactness are equivalent for $\{X_n\}$. Moreover, weak convergence is equivalent to tightness plus convergence of the finite-dimensional distributions. It is thus enough to prove that tightness implies (2.5) and that (2.5) implies tightness if the finite-dimensional distributions converge.

- Let us first assume that $X_n \xrightarrow{d} X$, hence tightness of the sequence $\{X_n\}$ and choose a compact set $B \subseteq C(K, S)$ such that $\sup_n \mathbb{P}(X_n \in B^c) < \varepsilon$. By the theorem of Arzelà-Ascoli we may choose $h > 0$ so small that $w(x, h) \leq \varepsilon$ for all $x \in B$. But then $\sup_n \mathbb{P}(w(X_n, h) > \varepsilon) < \varepsilon$, and (2.5) follows because ε was arbitrary.
- Next, let us assume convergence of the finite-dimensional distributions along with (2.5). Using that L^1 -convergence implies convergence in measure we may, for any $\varepsilon > 0$ choose a sequence $h_1, h_2, \dots > 0$ so small that

$$\sup_n \mathbb{P}(w(X_n, h_k) > 2^{-k}) \leq 2^{-(k+1)} \varepsilon, \quad k \in \mathbb{N}.$$

Next, let $\{t_n\} \subset K$ be dense. Convergence of the finite-dimensional distributions implies that the corresponding families of finite-dimensional distributions are tight. This allows us to choose compact sets $C_1, C_2, \dots \subseteq S$ such that

$$\sup_n \mathbb{P}(X_n(t_k) \in C_k^c) \leq 2^{-(k+1)} \varepsilon, \quad k \in \mathbb{N}.$$

Let us then define the set

$$B := \bigcap_{k \in \mathbb{N}} \left\{ x \in C(K, S) : x(t_k) \in C_k, w(x, h_k) \leq 2^{-k} \right\}.$$

Then, B is relatively compact, due to the Arzelà-Ascoli theorem. Moreover, $\sup_n \mathbb{P}(X_n \in B^c) \leq \varepsilon$.

□

Exercise 22 (Centsov Tightness Condition) (*) *Let $\{X^n\}$ be a sequence of continuous stochastic processes on $(\Omega, \mathcal{F}, \mathbb{P})$ that satisfy the following conditions:*

- $\sup_m \mathbb{E}[|X_0^m|^\nu] := M < \infty$.
- $\sup_m \mathbb{E}|X_t^m - X_s^m|^\alpha \leq C_T |t - s|^{1+\beta}$ for all $T > 0$ and $0 \leq s \leq t \leq T$

Follow the technique of the Kolmogorov-Centsov Theorem to prove that the sequence of probability measures $\{\mathbb{P} \circ (X^m)^{-1}\}$ is tight.

2.3.3 Donsker's invariance principle

Armed with the above results we can now to give an alternative proof of Donsker's invariance principle. As a byproduct we obtain a third construction of Brownian motion, through the weak convergence of a family of probability measures in $C([0, 1], \mathbb{R})$ to the Wiener measure.

To this end, we consider a sequence $(X_n)_{n \in \mathbb{N}}$ of $\mathbb{R}$ -valued i.i.d. random variables on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}(X_n) = 0$ and $\mathbb{E}(X_n^2) = 1$ and introduce the continuous time rescaled random walks

$$S_n(t) := \frac{1}{\sqrt{n}} \sum_{k=1}^{\lfloor nt \rfloor} X_k, \quad n \in \mathbb{N}$$

where $\lfloor x \rfloor := \max\{k \in \mathbb{Z} : k \leq x\}$ denotes the Gauss bracket of $x \in \mathbb{R}$. Let B be a Brownian motion. By the classical central limit theorem

$$(S_n(t_1) - S_n(t_0), \dots, S_n(t_N) - S_n(t_{N-1})) \xrightarrow{d} (B(t_1) - B(t_0), \dots, B(t_N) - B(t_{N-1})).$$

By linear interpolation we can then define the continuous process

$$S_n^*(t) := S_n(t) + \frac{tn - \lfloor tn \rfloor}{\sqrt{n}} X_{\lfloor tn \rfloor + 1}.$$

By Tschebyscheff's inequality

$$\mathbb{P}[|S_n^*(t) - S_n(t)| \geq \varepsilon] \rightarrow 0$$

and so it follows from Slutsky's theorem⁶ that

$$(S_n^*(t_1) - S_n^*(t_0), \dots, S_n^*(t_N) - S_n^*(t_{N-1})) \xrightarrow{d} (B(t_1) - B(t_0), \dots, B(t_N) - B(t_{N-1})).$$

As a result, the sequence of the linear interpolated rescaled random walks $\{S_n^*\}$ converges weakly to Brownian motion (i.e. the sequence of laws converges weakly to the Wiener measure) if we can prove that the sequence $\{S_n^*\}$ is tight. The argument uses the following maximum inequality for random walks.

Proposition 2.3.23 (Ottaviani) *Let $\{\xi_n\}$ be a sequence of i.i.d random variables with mean 0 and variance 1 and put $T_n := \sum_{i=1}^n \xi_i$. Let $\{T_n^*\}$ be the sequence of the running maxima of $\{|T_n|\}$. Then*

$$\mathbb{P}[T_n^* \geq 2r\sqrt{n}] \leq \frac{\mathbb{P}[|T_n| \geq r\sqrt{n}]}{1 - r^{-2}}, \quad r > 1, \quad n \in \mathbb{N}.$$

In particular,

$$\lim_{r \rightarrow \infty} r^2 \limsup_{n \rightarrow \infty} \mathbb{P}[T_n^* \geq r\sqrt{n}] = 0.$$

⁶We recall that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ does not imply that $X_n + Y_n \xrightarrow{d} X + Y$. Slutsky's theorem states that this is true if Y is a deterministic constant.

PROOF: Let $A_k := \{T_{k-1}^* \leq 2r, |T_k| > 2r\}$ the event where the random walk exceeds the level $2r$ for the first time at time $k \in \mathbb{N}$. Then the following holds:

$$\begin{aligned}
\mathbb{P}[|T_n| > r] &= \sum_{k=1}^n \mathbb{P}(|T_n| > r, A_k) + \mathbb{P}(|T_n| > r, A_n^c) \\
&\geq \sum_{k=1}^n \mathbb{P}(|T_n| > r, A_k) \\
&\geq \sum_{k=1}^n \mathbb{P}(|T_n - T_k| \leq r, A_k) \\
&\geq \min_{1 \leq k \leq n} \mathbb{P}(|T_n - T_k| \leq r) \mathbb{P}(\cup_{k=1}^n A_k) \quad \text{- independence of } \{|T_n - T_k| \leq r\}, A_k \\
&= \min_{1 \leq k \leq n} \mathbb{P}(|T_k| \leq r) \mathbb{P}(T_n^* > 2r).
\end{aligned}$$

Let $c := r\sqrt{n}$. Then it follows from Chebyshev's inequality that

$$\min_{1 \leq k \leq n} \mathbb{P}[|T_k| \leq c] \geq \min_{1 \leq k \leq n} (1 - kc^{-2}) \geq (1 - r^{-2}).$$

This shows the first assertion. The second assertion follows from the central limit theorem along with the Portmanteau theorem and the already established fact (see proof of Lemma [\(2.2.2\)](#)) that for any $N(0, 1)$ random variable Z and $r > 1$,

$$\mathbb{P}[|Z| > r] \leq \sqrt{\frac{2}{\pi}} e^{-r^2/2}.$$

□

Theorem 2.3.24 (Donsker's invariance principle) *Let B is a Brownian motion. Under the above assumption, restricted to any compact interval (and hence on $[0, \infty)$)*

$$S_n^* \xrightarrow{d} B.$$

PROOF: The statement follows as soon as we can prove the tightness of the sequence $\{S_n^*\}$. For this we apply the previous result to the sequence

$$T_{[nt]} = \sqrt{n} S_n(t) \quad \text{so that} \quad T_{[nh]}^* = \sqrt{n} \sup_{0 \leq r \leq h} |S_n(r)|.$$

In particular,

$$T_{[nh]}^* \geq r\sqrt{[nh]} \Leftrightarrow \sup_{0 \leq r \leq h} S_n(r) \geq r\sqrt{\frac{[nh]}{n}} \approx r\sqrt{h}.$$

We hence obtain from Ottaviani's lemma that

$$\lim_{r \rightarrow \infty} r^2 \lim_{n \rightarrow \infty} \mathbb{P}[\sup_{0 \leq t \leq h} |S_n(t)| \geq r\sqrt{h}] = 0.$$

Setting $r = \varepsilon(\sqrt{h})^{-1}$ and using the independence of the increments of the sums, this shows that

$$\lim_{h \rightarrow 0} h^{-1} \limsup_{n \rightarrow \infty} \sup_{t \in [0, 1]} \mathbb{P} \left[\sup_{0 \leq r \leq h} |S_n^*(t+r) - S_n^*(t)| > \varepsilon \right] = 0.$$

Now (2.5) follows by dividing $[0, 1]$ into h^{-1} subintervals $[t_{k-1}, t_k]$ of length h :

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \mathbb{P} \left[\sup_{0 \leq t \leq 1, r \leq h} |S_n^*(t+r) - S_n^*(t)| > \varepsilon \right] \\ & \leq \sum_{k=1}^{h^{-1}} \limsup_{n \rightarrow \infty} \mathbb{P} \left[\sup_{r \leq h} |S_n^*(t_{k-1}+r) - S_n^*(t_{k-1})| > \varepsilon \right] \\ & = h^{-1} \limsup_{n \rightarrow \infty} \sup_{t \in [0, 1]} \mathbb{P} \left[\sup_{0 \leq r \leq h} |S_n^*(t+r) - S_n^*(t)| > \varepsilon \right] \rightarrow 0. \end{aligned}$$

□

2.4 The Brownian filtration

The natural filtration $\mathcal{F}^X$ of a right-continuous process is not right-continuous in general. A simple counter example is given by the process

$$X_t = tZ, \quad t \geq 0$$

where Z is some non-trivial random variable. In this case $\mathcal{F}_0^X = \{\emptyset, \Omega\}$ while $\mathcal{F}_\varepsilon^X = \sigma(Z)$ for all $\varepsilon > 0$. Hence

$$\mathcal{F}_{0+}^X \neq \mathcal{F}_0^X.$$

However it can be shown that the completion of the natural filtration of a Feller Markov process, including all Lévy processes (stochastically continuous càdlàg processes starting in zero that have stationary, independent increments) and in particular Brownian motion, is always right-continuous.

In what follows we prove this result for Brownian motion. To this end, we need a slight modification of the definition of the Brownian motion that includes a filtration.

Definition 2.4.1 *Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be a filtered probability space. An adapted process B is called a Brownian motion w.r.t. the filtration $(\mathcal{F}_t)$ if*

- $B_0 = 0$ a.s.
- $B_t - B_s \perp \mathcal{F}_s$ for all $0 \leq s \leq t < \infty$
- $B_t - B_s \sim N(0, t-s)$
- B has almost surely continuous sample paths

Exercise 23 *Let B be a Brownian motion in the sense of the original definition and let*

$$\mathcal{F}_t^B := \sigma(B_s : 0 \leq s \leq t)$$

be the filtration generated by B . Then B is a Brownian motion w.r.t. the filtration $(\mathcal{F}_t^B)$.

The filtration $\mathcal{F}^B$ generated by a Brownian motion B is not right-continuous. Its completion

$$\mathcal{F}_t := \mathcal{F}_t^B \vee \mathcal{N} \tag{2.6}$$

however, is, as shown by the next result.

Remark 2.4.2 We notice that a Brownian motion remains a BM with respect to the completed filtration: completion of σ -fields does not affect the assumptions of Definition [2.4.1](#).

To prove the next result, we recall that by the monotone class theorem, if some property holds for all elements from a $\cap$ -stable generator of a σ -field, then it holds for all elements from the σ -field. Since $\mathcal{F}^B$ is generated by the projection mappings it is the smallest σ -field with respect to which all bounded functions depending on finitely many values $B_{t_1}, \dots, B_{t_d}$ are measurable. Hence, if we can prove that a property holds for all bounded measurable functions depending only on finitely many coordinates, then it holds for all events in $\mathcal{F}^B$.

Proposition 2.4.3 The filtration $(\mathcal{F}_t)$ introduced in [\(2.6\)](#) satisfies the usual conditions. In particular, it is right-continuous.

PROOF: Completeness is clear. To prove the right-continuity we fix $d \in \mathbb{N}$, $0 \leq t_1 < \dots < t_d$ and a bounded and continuous function $f: \mathbb{R}^d \rightarrow \mathbb{R}$. We claim that the conditional expectation

$$\mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+}]$$

is $\mathcal{F}_t$ -measurable. Indeed, let

$$t_k < t < t + \frac{1}{n} < t_{k+1}$$

and $p_n(x_1, \dots, x_{d-k})$ be the density of the increments $(B_{t_{k+1}} - B_{t+1/n}, \dots, B_{t_d} - B_{t+1/n})$. Then,

$$\begin{aligned} & \mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+1/n}] \\ &= \int_{\mathbb{R}^{d-k}} f(B_{t_1}, \dots, B_{t_k}, B_{t+1/n} + x_1, \dots, B_{t+1/n} + x_{d-k}) p_n(x_1, \dots, x_{d-k}) dx_1 \dots dx_{d-k}. \end{aligned}$$

The densities p_n converge to the density p of $(B_{t_{k+1}} - B_t, \dots, B_{t_d} - B_t)$. By dominated convergence, using the continuity of f and B ,

$$\begin{aligned} & \mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+1/n}] \\ & \rightarrow \int_{\mathbb{R}^{d-k}} f(B_{t_1}, \dots, B_{t_k}, B_t + x_1, \dots, B_t + x_{d-k}) p(x_1, \dots, x_{d-k}) dx_1 \dots dx_{d-k}, \end{aligned}$$

which is $\mathcal{F}_t$ -measurable for any $n \in \mathbb{N}$. At the same time, by Lemma [1.3.15](#) (backwards martingale convergence)

$$\mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+1/n}] \rightarrow \mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+}] \quad \text{a.s.}$$

Since $\mathcal{N} \in \mathcal{F}_t$ we see that $\mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+}]$ is indeed $\mathcal{F}_t$ -measurable. The reason we need completeness is that we only have a.s. convergence.⁷ Since for any open interval (a, b) the indicator function $\mathbf{1}_{(a,b)}$ can be approximated pointwise by an increasing sequence of continuous functions it follows from dominated convergence that the $\mathcal{F}_t$ -measurability carries over to all functions of the form

$$f = \mathbf{1}_{(a_1, b_1) \times \dots \times (a_d, b_d)}.$$

Hence, it follows from the monotone class theorem that

$$\mathbb{E}[\mathbf{1}_A | \mathcal{F}_{t+}] \quad \text{is } \mathcal{F}_t\text{-measurable}$$

⁷We recall that an a.s. limit of a sequence of measurable functions is measurable only if the underlying filtration is complete!

for any $A \in \mathcal{F}^B$. In particular, for $A \in \mathcal{F}_{t+}$,

$$\mathbf{1}_A = \mathbb{E}[\mathbf{1}_A | \mathcal{F}_{t+}] \quad \text{is } \mathcal{F}_t\text{-measurable.}$$

Hence $\mathcal{F}_{t+} = \mathcal{F}_t$. □

2.5 Sample path properties

In this section we establish two important sample path properties of Brownian motion: the nowhere differentiability and the law of iterated logarithm.

2.5.1 Nowhere differentiability

By construction almost every Brownian sample path is α -Hölder continuous for $\alpha \in (0, 1/2)$. We now show that a typical Brownian path is nowhere differentiable. To this end, we recall that the upper and lower right Dini derivatives of a continuous function $f : [0, \infty) \rightarrow \mathbb{R}$ are given by

$$D^+ f(t) := \limsup_{h \downarrow 0} \frac{f(t+h) - f(t)}{h} \quad \text{and} \quad D_+ f(t) := \liminf_{h \downarrow 0} \frac{f(t+h) - f(t)}{h}.$$

Theorem 2.5.1 *Let B be a standard Brownian motion. A typical sample path $B(\omega)$ is nowhere differentiable. More precisely, the set*

$$\{\omega \in \Omega : \text{for each } t \in [0, \infty) \text{ either } D^+ B_t(\omega) = \infty \text{ or } D_+ B_t(\omega) = -\infty\}$$

contains an event F with $\mathbb{P}(F) = 1$.

PROOF: It is enough to consider the time interval $[0, 1]$. For fixed integers $j, k \geq 1$ we define the set

$$A_{j,k} = \bigcup_{t \in [0,1]} \bigcap_{h \in [0,1/k]} \{\omega \in \Omega : |B_{t+h}(\omega) - B_t(\omega)| \leq jh\}.$$

The union of the sets $A_{j,k}$ collects all those ω for which the path $t \mapsto B_t(\omega)$ has a locally Lipschitz continuous increment somewhere in $[0, 1]$. In particular,

$$\{\omega \in \Omega : -\infty < D_+ B_t(\omega) \leq D^+ B_t(\omega) < \infty \text{ for some } t \in [0, 1]\} \subset \bigcup_{j=1}^{\infty} \bigcup_{k=1}^{\infty} A_{j,k}.$$

We will show that for any pair (j, k) there exists a set C of $\mathbb{P}$ -measure zero such that $A_{j,k} \subseteq C$. Since there are only countably many pairs $(j, k) \in \mathbb{N} \times \mathbb{N}$ this shows that there exists a set N of measure zero, such that

$$\{\omega \in \Omega : -\infty < D_+ B_t(\omega) \leq D^+ B_t(\omega) < \infty \text{ for some } t \in [0, 1]\} \subseteq N.$$

Let us fix a sample path $\omega \in A_{j,k}$. Our first goal is to show that the assumption $\omega \in A_{j,k}$ provides information about the size of the Brownian increments over *neighboring* intervals of the same length.

Indeed, by the definition of the set $A_{j,k}$ there exists $t \in [0, 1)$ such that

$$|B_{t+h}(\omega) - B_t(\omega)| \leq jh \quad \text{for all } h \in [0, 1/k]. \quad (2.7)$$

We now we fix an integer $n \geq 4k$ and divide the interval $[0, 1]$ into equi-distant subintervals of length $\frac{1}{n}$. Then $\frac{i-1}{n} \leq t \leq \frac{i}{n}$ for some integer $1 \leq i \leq n$, and using that $n \geq 4k$,

$$\frac{i+v}{n} - t \leq \frac{v+1}{n} \leq \frac{1}{k} \quad \text{for } v = 1, 2, 3. \quad (2.8)$$

As a result, the quantities

$$h_1 := \frac{i}{n} - t \leq \frac{1}{n} \leq \frac{1}{k}, \quad \text{and} \quad h_2 := \frac{i+1}{n} - t \leq \frac{2}{n} \leq \frac{1}{k}$$

satisfy $h_1, h_2 \leq \frac{1}{k}$ and are hence “admissible” in equation (2.7). Using that $h_2 \leq 2/n$ and $h_1 \leq 1/n$ we see that the increment of the Brownian motion on the interval that contains t satisfies

$$|B_{(i+1)/n}(\omega) - B_{i/n}(\omega)| \leq |B_{(i+1)/n}(\omega) - B_t(\omega)| + |B_{i/n}(\omega) - B_t(\omega)| \leq \frac{2j}{n} + \frac{j}{n} = \frac{3j}{n}.$$

The assumption $\omega \in A_{j,k}$ also provides information about the size of the Brownian increment over the next two intervals. Applying (2.8) and (2.7) to h_2 and

$$h_3 := \frac{i+2}{n} - t \leq \frac{3}{n} \leq \frac{1}{k}$$

yields

$$\begin{aligned} |B_{(i+2)/n}(\omega) - B_{(i+1)/n}(\omega)| &\leq |B_{(i+2)/n}(\omega) - B_t(\omega)| + |B_{(i+1)/n}(\omega) - B_t(\omega)| \\ &\leq \frac{3j}{n} + \frac{2j}{n} = \frac{5j}{n} \end{aligned}$$

and applying it to h_3 and

$$h_4 := \frac{i+3}{n} - t \leq \frac{4}{n} \leq \frac{1}{k}$$

yields

$$\begin{aligned} |B_{(i+3)/n}(\omega) - B_{(i+2)/n}(\omega)| &\leq |B_{(i+3)/n}(\omega) - B_t(\omega)| + |B_{(i+2)/n}(\omega) - B_t(\omega)| \\ &\leq \frac{4j}{n} + \frac{3j}{n} = \frac{7j}{n}. \end{aligned}$$

With

$$C_i^{(n)} := \bigcap_{v=1}^3 \left\{ \omega \in \Omega : |B_{(i+v)/n}(\omega) - B_{(i+v-1)/n}(\omega)| \leq \frac{2v+1}{n} j \right\} \quad (2.9)$$

we therefore conclude that $A_{j,k} \subset \bigcup_{i=1}^n C_i^{(n)}$, for all $n \geq 4k$. Moreover, the random variables

$$\sqrt{n}(B_{(i+v)/n} - B_{(i+v-1)/n}) =: Z_v \quad (v = 1, 2, 3)$$

are independent and $\mathcal{N}(0, 1)$ distributed. Since

$$\mathbb{P}(|Z_v| \leq \varepsilon) \leq \varepsilon$$

it follows from (2.9) that

$$\mathbb{P}[C_i^{(n)}] \leq \frac{105j^3}{n^{3/2}} \quad \left(105 = \prod_{\nu=1}^3 (2\nu+1) \right)$$

and hence that $\mathbb{P}[\cup_{i=1}^n C_i^{(n)}] = \mathcal{O}(n^{-1/2})$. Since

$$A_{j,k} \subseteq C := \bigcap_{n=4k}^{\infty} \bigcup_{i=1}^n C_i^{(n)}$$

the assertion follows from

$$\mathbb{P}[C] \leq \inf_{n \geq 4k} \mathbb{P}[\cup_{i=1}^n C_i^{(n)}] = 0.$$

□

Exercise 24 Prove that a Brownian sample path is a.s. nowhere Hölder continuous with exponent $\alpha > \frac{1}{2}$.

Hint. Follow the proof of nowhere differentiability.

2.5.2 The law of iterated logarithm

Brownian sample paths are unbounded. The law of iterated logarithm gives the exact growth rate of Brownian motion at 0 and ∞ . It requires the following auxiliary result on the tail of the normal distribution.

Lemma 2.5.2 Let $X \sim N(0, 1)$. For every $x > 0$, we have

$$\frac{1}{\sqrt{2\pi}} \frac{1}{x + 1/x} e^{-x^2/2} \leq \mathbb{P}(X \geq x) \leq \frac{1}{\sqrt{2\pi}} \frac{1}{x} e^{-x^2/2}.$$

PROOF: Let φ be the density of the standard normal distribution. By partial integration ($u = 1/t, v' = t\varphi(t)$)

$$\mathbb{P}(X \geq x) = \int_x^{\infty} \frac{1}{t} (t\varphi(t)) dt = -\frac{1}{t} \varphi(t) \Big|_x^{\infty} - \int_x^{\infty} \frac{1}{t^2} \varphi(t) dt \leq \frac{1}{x} \varphi(x).$$

By analogy

$$\mathbb{P}(X \geq x) \geq \frac{1}{x} \varphi(x) - \frac{1}{x^2} \int_x^{\infty} \varphi(t) dt = \frac{1}{x} \varphi(x) - \frac{1}{x^2} \mathbb{P}(X \geq x),$$

from which we deduce the second inequality. □

Theorem 2.5.3 (Law of iterated logarithm) For a Brownian motion B in $\mathbb{R}$ we have a.s.

$$\limsup_{t \rightarrow 0} \frac{B_t}{\sqrt{2t \log \log(1/t)}} = \limsup_{t \rightarrow \infty} \frac{B_t}{\sqrt{2t \log \log t}} = 1$$

PROOF: The inversion $t \mapsto tB_{1/t}$ converts the two formulas onto one another, so we only need to prove the result for $t \rightarrow \infty$.

- To prove “ $\leq$ ” we set $t_n = \alpha^n$ for some $\alpha > 1$ (we later let $\alpha \downarrow 1$) and put

$$f(t) := 2\alpha^2 \log \log t.$$

Then

$$\frac{f(t_n)}{2\alpha} = \alpha (\log(n \log \alpha)) = \alpha \log n + \alpha \log \log \alpha.$$

Let us also put

$$B_{a,b} = \{B_t\}_{a \leq t \leq b}$$

and denote by B^* the running maximum of the Brownian motion. Since $\mathbb{P}[B_t^* > a] = 2\mathbb{P}[B_t > a]$ it follows from the above lemma that up to a multiplicative constant

$$\begin{aligned} \mathbb{P}\left(\sup_{t_n, t_{n+1}} B_{t_n, t_{n+1}} > \sqrt{t_n f(t_n)}\right) &\leq \mathbb{P}\left(t_{n+1}^{-1/2} B_{t_{n+1}}^* > \sqrt{f(t_n)/\alpha}\right) \quad (\text{using that } t_{n+1} = \alpha t_n) \\ &= \mathbb{P}\left(B_1^* > \sqrt{f(t_n)/\alpha}\right) \\ &\leq \sqrt{\frac{\alpha}{f(t_n)}} e^{-f(t_n)/2\alpha} \\ &= (\log \alpha)^{-\alpha} \sqrt{\frac{\alpha}{f(t_n)}} n^{-\alpha} \\ &\leq n^{-\alpha} \quad \text{for } n \text{ large enough.} \end{aligned}$$

Since $\alpha > 1$, it follows from the Borel-Cantelli lemma that

$$\mathbb{P}\left(\sup_{t_n, t_{n+1}} B_{t_n, t_{n+1}} > \sqrt{t_n f(t_n)} \text{ infinitely often}\right) = 0.$$

Since the function $t \mapsto \sqrt{t f(t)}$ is non-decreasing this shows that a.s. for all $n \geq n^*(\omega)$

$$\sup_{t_n \leq t \leq t_{n+1}} \frac{B_t}{\sqrt{t f(t)}} \leq \sup_{t_n \leq t \leq t_{n+1}} \frac{B_t}{\sqrt{t_n f(t_n)}} \leq 1$$

and hence that

$$\limsup_{t \rightarrow \infty} \frac{B_t}{\sqrt{t f(t)}} \leq 1 \quad \text{a.s.}$$

Letting $\alpha \downarrow 1$ yields

$$\limsup_{t \rightarrow \infty} \frac{B_t}{\sqrt{2t \log \log t}} \leq 1.$$

- To prove “ $\geq$ ” we consider again the sequence $\{t_n\}$ and prove that for $\eta > 0$ we can find a large enough α such that

$$\limsup_{n \rightarrow \infty} \frac{B_{t_n}}{\sqrt{2t_n \log \log t_n}} \geq 1 - \eta \quad \text{a.s.} \quad (2.10)$$

We first note that the previously obtained upper bound applied to the Brownian motion $-B$ yields

$$\liminf_{t \rightarrow \infty} \frac{B_t}{\sqrt{2t \log \log t}} \geq -1 \quad (2.11)$$

and set

$$\beta := \frac{\alpha}{\alpha - 1}, \quad g(t) := \frac{2}{\beta^2} \log \log t.$$

We may assume w.l.o.g. that $\beta g(t_n) \geq 1$ and note that

$$e^{-\beta g(t_n)/2} = n^{-1/\beta} (\log \alpha)^{-1/\beta}, \quad \frac{t_n g(t_n)}{2t_n \log \log t_n} = \beta^{-2}. \quad (2.12)$$

Since $t_n - t_{n-1} = \beta t_n$ it follows from the scaling property of Brownian motion that

$$\mathbb{P}\left(B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)}\right) = \mathbb{P}\left(B_1 > \sqrt{\beta g(t_n)}\right)$$

We now apply the lower bound in Lemma 2.5.2 to $x = \sqrt{\beta g(t_n)} \geq 1$. Since $(x + 1/x)^{-1} \geq \frac{1}{2x}$ for $x \geq 1$ we get that

$$\begin{aligned} \mathbb{P}\left(B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)}\right) &= \mathbb{P}\left(B_1 > \sqrt{\beta g(t_n)}\right) \\ &\geq \frac{1}{\sqrt{2\pi}} \frac{1}{2} \frac{1}{\sqrt{\beta g(t_n)}} e^{-\beta g(t_n)/2} \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{2} (\log \alpha)^{-1/\beta} \sqrt{\frac{1}{\beta g(t_n)}} n^{-1/\beta} \\ &\geq n^{-\varepsilon} n^{-1/\beta} \\ &\geq n^{-1} \end{aligned}$$

for $\varepsilon \in (0, 1 - 1/\beta)$ and $n \in \mathbb{N}$ large enough. Since the events

$$\{B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)}\}_{n \in \mathbb{N}}$$

are independent (this is important!) it follows from the Borel-Cantelli lemma that

$$\mathbb{P}\left(B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)} \text{ infinitely often}\right) = 1.$$

This allows us to bound first $B_{t_{n-1}}$ and then $\frac{B_{t_n}}{\sqrt{2t_n \log \log t_n}}$ from below using (2.11) and the fact that

$$\frac{t_n \log \log t_n}{t_{n-1} \log \log t_{n-1}} \rightarrow \alpha.$$

In fact, by (2.11)

$$B_{t_{n-1}} > -(1 + \varepsilon) \alpha^{-1/2} \sqrt{2t_n \log \log t_n} \quad \text{for almost all } n \in \mathbb{N} \text{ a.s.}$$

and by the second equation in (2.12)

$$\limsup_{n \rightarrow \infty} \frac{B_{t_n}}{\sqrt{2t_n \log \log t_n}} \geq \frac{1}{\beta} - (1 + \varepsilon) \alpha^{-1/2} = \frac{\alpha - 1}{\alpha} - (1 + \varepsilon) \alpha^{-1/2}.$$

Choosing first α large and then ε small we obtain (2.10). This prove the desired lower bound. $\square$

2.5.3 Nowhere monotonicity

Next, we prove that a typical Brownian sample path (starting in zero) changes its sign infinitely often on any small time interval. From this we conclude that the Brownian sample paths are almost surely nowhere monotone. Our first result is Blumenthal's 0-1 law.

Proposition 2.5.4 (Blumenthal's 0-1 law) *Let B be a Brownian motion and $(\mathcal{F}_t)$ be its canonical filtration. The σ -field $\mathcal{F}_{0+} := \bigcap_{t>0} \mathcal{F}_t$ is trivial in the sense that*

$$\mathbb{P}(A) \in \{0, 1\} \quad \text{for all } A \in \mathcal{F}_{0+}.$$

PROOF: Let $A \in \mathcal{F}_{0+}$, let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and continuous and let $0 < t_1 < \dots < t_d$. By continuity and monotone convergence

$$\mathbb{E}[\mathbf{1}_A f(B_{t_1}, \dots, B_{t_d})] = \lim_{\varepsilon \rightarrow 0} \mathbb{E}[\mathbf{1}_A f(B_{t_1} - B_\varepsilon, \dots, B_{t_d} - B_\varepsilon)].$$

If $\varepsilon < t_1$, then the above increments are independent of $\mathcal{F}_\varepsilon$ and hence of $\mathcal{F}_{0+}$. Thus,

$$\mathbb{E}[\mathbf{1}_A f(B_{t_1}, \dots, B_{t_d})] = \lim_{\varepsilon \rightarrow 0} \mathbb{E}[\mathbf{1}_A f(B_{t_1} - B_\varepsilon, \dots, B_{t_d} - B_\varepsilon)] = \mathbb{P}(A) \mathbb{E}[f(B_{t_1}, \dots, B_{t_d})].$$

This shows that $\mathcal{F}_{0+} \perp \sigma(B_{t_1}, \dots, B_{t_d})$ and hence

$$\mathcal{F}_{0+} \perp \sigma(B_t, t > 0).$$

Since B_0 is deterministic,

$$\sigma(B_t, t > 0) = \sigma(B_t, t \geq 0)$$

and so $\mathcal{F}_{0+} \perp \mathcal{G}$ for any sub- σ -field $\mathcal{G} \subset \sigma(B_t, t \geq 0)$. Choosing $\mathcal{G} = \mathcal{F}_{0+}$ we see that

$$\mathcal{F}_{0+} \perp \mathcal{F}_{0+}$$

and hence $\mathcal{F}_{0+}$ is trivial. □

Corollary 2.5.5 *For any $\varepsilon > 0$,*

$$\sup_{0 \leq s \leq \varepsilon} B_s > 0, \quad \inf_{0 \leq s \leq \varepsilon} B_s < 0.$$

PROOF: Let $\varepsilon_n \downarrow 0$ and set $A := \bigcap_n \{\sup_{0 \leq s \leq \varepsilon_n} B_s > 0\}$. Since this is a monotone decreasing intersection, $A \in \mathcal{F}_{0+}$ and

$$\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(\sup_{0 \leq s \leq \varepsilon_n} B_s > 0).$$

At the same time, by the symmetry of Brownian motion

$$\mathbb{P}(\sup_{0 \leq s \leq \varepsilon_n} B_s > 0) \geq \mathbb{P}(B_{\varepsilon_n} > 0) = \frac{1}{2},$$

hence $\mathbb{P}(A) > 0$. Therefore, by Blumenthal's 0-1 law, $\mathbb{P}(A) = 1$. The proof for the infimum is similar. □

It follows from the above corollary that Brownian sample paths are almost surely nowhere monotone. In particular, they cannot be viewed as (random) distribution functions. Later, we show that Brownian sample paths also cannot be represented as the difference of monotone functions. Hence, we cannot integrate w.r.t. Brownian sample paths in the sense of real analysis.